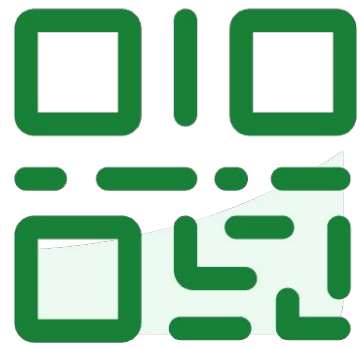


Do not edit
How to change the
design



**Join at slido.com
#2474272**

① The Slido app must be installed on every computer you're presenting from

slido

Course Evaluation



The final course survey is intended to help us improve future offerings of the course.

If you complete our **internal anonymous survey AND the official Course Evaluation** (<http://course-evaluations.berkeley.edu>) you will get **0.5% extra credit points** to your grade. If **90%** of the class also submits both, then everyone who **submitted both** will receive **an additional 0.5% percentage point** on their final grade.



Internal course survey link:
<https://bit.ly/cs189-eval>

LLM for Specific Tasks



July 21, 2025 Research

Advanced version of Gemini with Deep Think officially achieves gold-medal standard at the International Mathematical Olympiad

Thang Luong and Edward Lockhart

Share 



Catalin 
@catalinmpit

Next generation of developers.

```
2  
3  
4   a = 5  
5   b = 3  
6  
7   sum = OpenAI.chat("Sum of #{a} + #{b}")  
8  
9   print(sum)  
10  
11
```

LLM for Specific Tasks



- **Supervised Fine Tuning (SFT):**
Finetune a pre-trained model on a smaller, labeled dataset for a specific task
- **How do we achieve this without losing the ability to generalize?**



LLM for Specific Tasks



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- **Supervised Fine Tuning (SFT):**
Finetune a pre-trained model on a smaller, labeled dataset for a specific task
- **How do we achieve this without losing the ability to generalize?**

How Abilities in Large Language Models are Affected by Supervised Fine-tuning Data Composition

Guanting Dong*, Hongyi Yuan*, Keming Lu, Chengpeng Li*, Mingfeng Xue
Dayiheng Liu, Wei Wang, Zheng Yuan, Chang Zhou, Jingren Zhou
Alibaba Group
{dongguanting.dgt,yuanzheng.yuanzhen,ericzhou.zc}@alibaba-inc.com

Abstract

Large language models (LLMs) with enormous pre-training tokens and parameters emerge diverse abilities, including math reasoning, code generation, and instruction following. These abilities are further enhanced by supervised fine-tuning (SFT). While the open-source community has explored ad-hoc SFT for enhancing individual capabilities, proprietary LLMs exhibit versatility across various skills. Therefore, understanding the facilitation of multiple abilities via SFT is paramount. In this study, we

tasks expressed in natural languages (Ouyang et al., 2022a; Anil et al., 2023; OpenAI, 2023; Luo et al., 2023a), especially Information Extraction (IE) (Lu et al., 2022; Xu et al., 2023b; Zhao et al., 2023; Cheng et al., 2023d; Wang et al., 2023b; Zhang et al., 2024b; Li et al., 2024), Information Retrieval (IR) (Zhu et al., 2024; Liu et al., 2024c) and Spoken Language Understanding (SLU) (Hosilowicz et al., 2024; Yin et al., 2024; Cheng et al., 2023a, 2024; Dong et al., 2023a). Among the tasks, LLMs especially emerge with three outstanding abilities



TERRY KIM · COMMUNITY PREDICTION COMPETITION · PRIVATE · 18 DAYS TO GO

CS189-HW5 (LLM SFT)

Answering MCQ with Fine-Tuning Challenge

Overview Data Code Models Discussion Leaderboard Rules Team Submissions Settings

Overview

Answering MCQ with Fine-Tuning Challenge

Welcome to the final homework! Phew. This mini-competition is designed to help you learn supervised fine-tuning (SFT) by applying it to a question-answering task based on mix of different types of questions.

Start

2 days ago



Close

18 days to go

Homework 5!

Due Date: December 12

- Paper Question
- Kaggle Submission (With Write-up)



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Important!

- 3 Submissions; Paper Questions, Code [Notebook + PDF], Writeup
- Make sure you submit with proper format
- You can submit up to 5 Kaggle submissions per day

Lecture 25

Contrastive Learning and Diffusion Models

Contrastive representation learning and diffusion models

EECS 189/289, Fall 2025 @ UC Berkeley

Joseph E. Gonzalez and Narges Norouzi

Slides adopted from Steve Seitz,
Andrew Owens, etc.

Roadmap

- Discriminative
 - Rotation
 - Jigsaw
 - Clustering
 - Contrastive Learning
- Diffusion Models



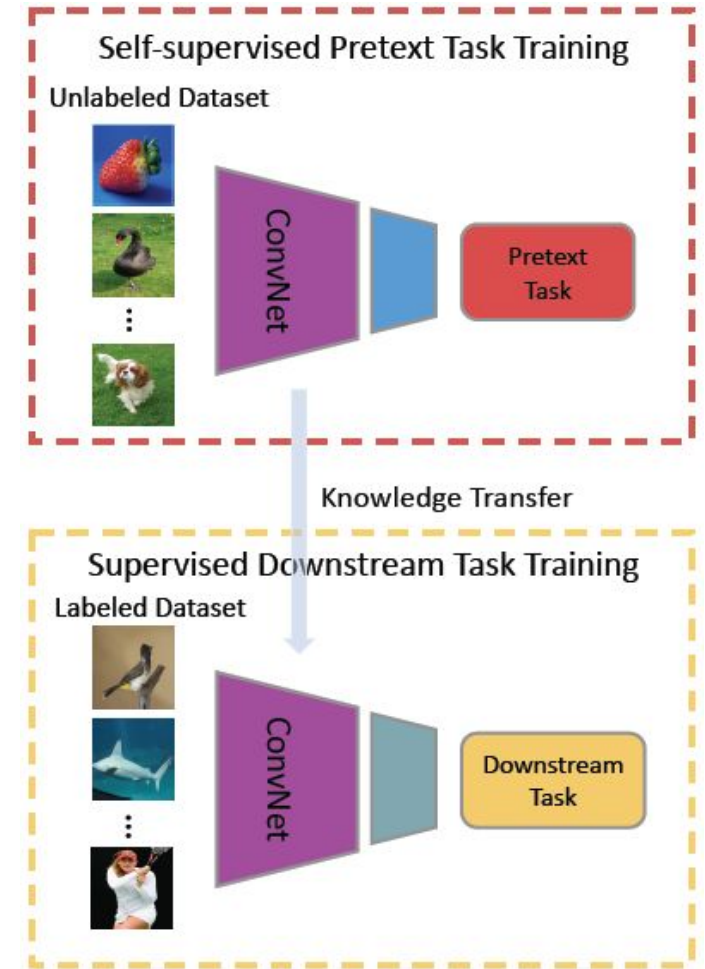
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Self-Supervised Learning



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- **Unsupervised Learning:** Model isn't told what to predict.
- **Self-Supervised Learning:** Representation learning with **unlabeled data**
 - Learn useful **feature representations** from unlabeled data through **pre-text tasks**.
 - The term “self-supervised” refers to creating **its own supervision** (i.e., without supervision, without labels)



Contrastive Representation Learning

- Discriminative
 - Rotation
 - Jigsaw
 - Clustering
 - Contrastive Learning
- Diffusion Models

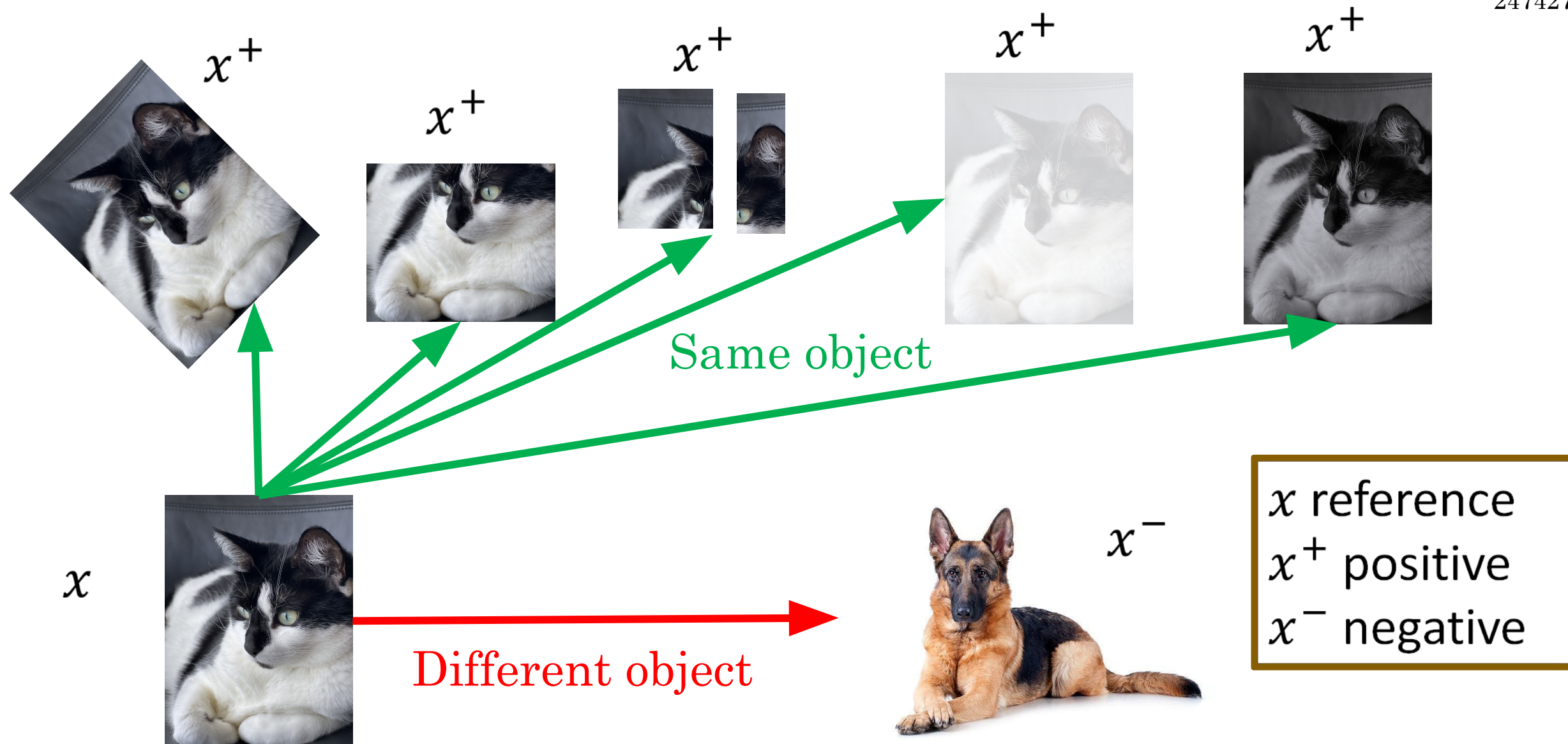


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Contrastive Representation Learning



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Formulation for Contrastive Learning



- What we want:

$$\textit{score}(f(x), f(x^+)) \gg \textit{score}(f(x), f(x^-))$$

x reference
 x^+ positive
 x^- negative

- Given a chosen score function, we aim to **learn an encoder function f** that yields high score for positive pairs (x, x^+) and low scores for negative pairs (x, x^-) .

InfoNCE Loss



- InfoNCE (Information Noise-Contrastive Estimation) loss minimizes **the similarity** between an anchor sample and a set of negative (unrelated) samples.
- Given 1 positive sample and N-1 negative samples:

$$L = -\mathbb{E}_x \left[\log \frac{e^{s(f(x), f(x^+))}}{e^{s(f(x), f(x^+))} + \sum_{j=1}^{N-1} e^{s(f(x), f(x^-))}} \right]$$

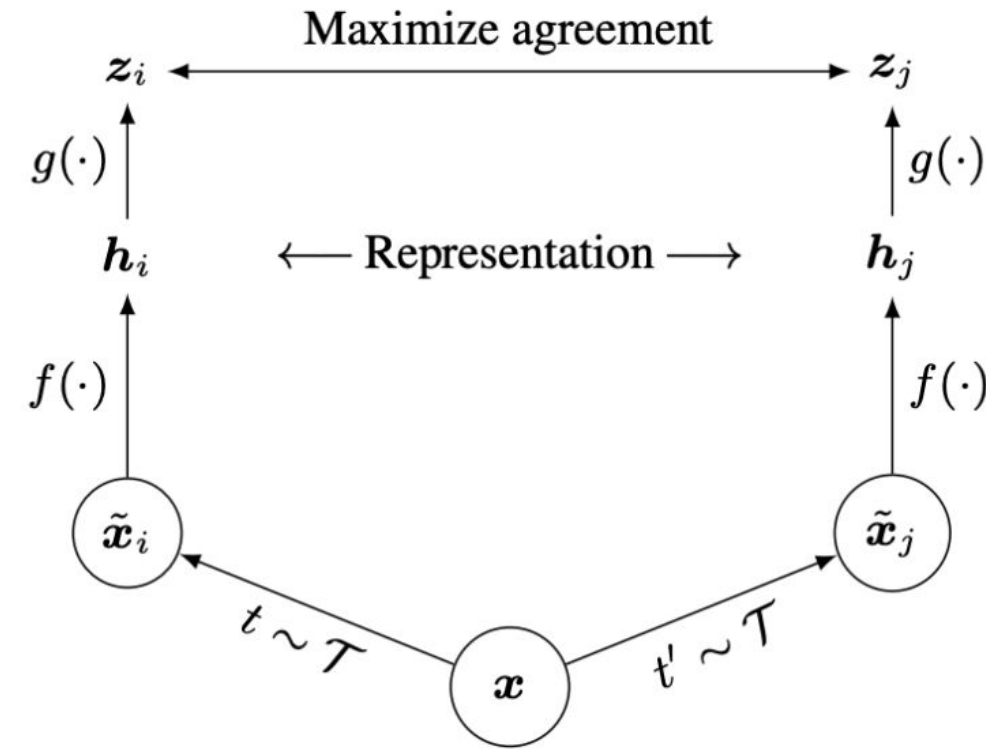
This is similar to cross-entropy loss for a N-way softmax classifier.

SimCLR: A Simple Framework for Contrastive Learning

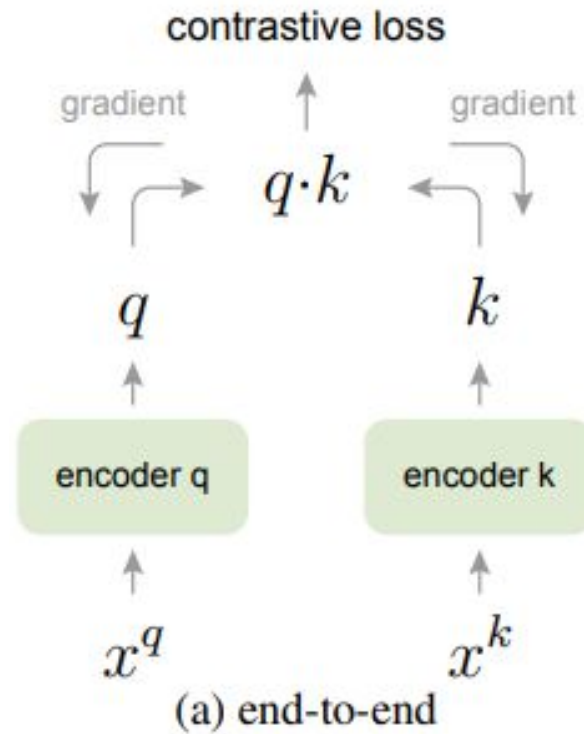


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- Cosine similarity as the score function
- Use a projection network $g(\cdot)$ to project features to a space where contrastive learning is applied.
- Generate positive samples through data augmentation:
 - Random cropping, random color distortion, and random blur.



Momentum Contrastive Learning (MoCo)



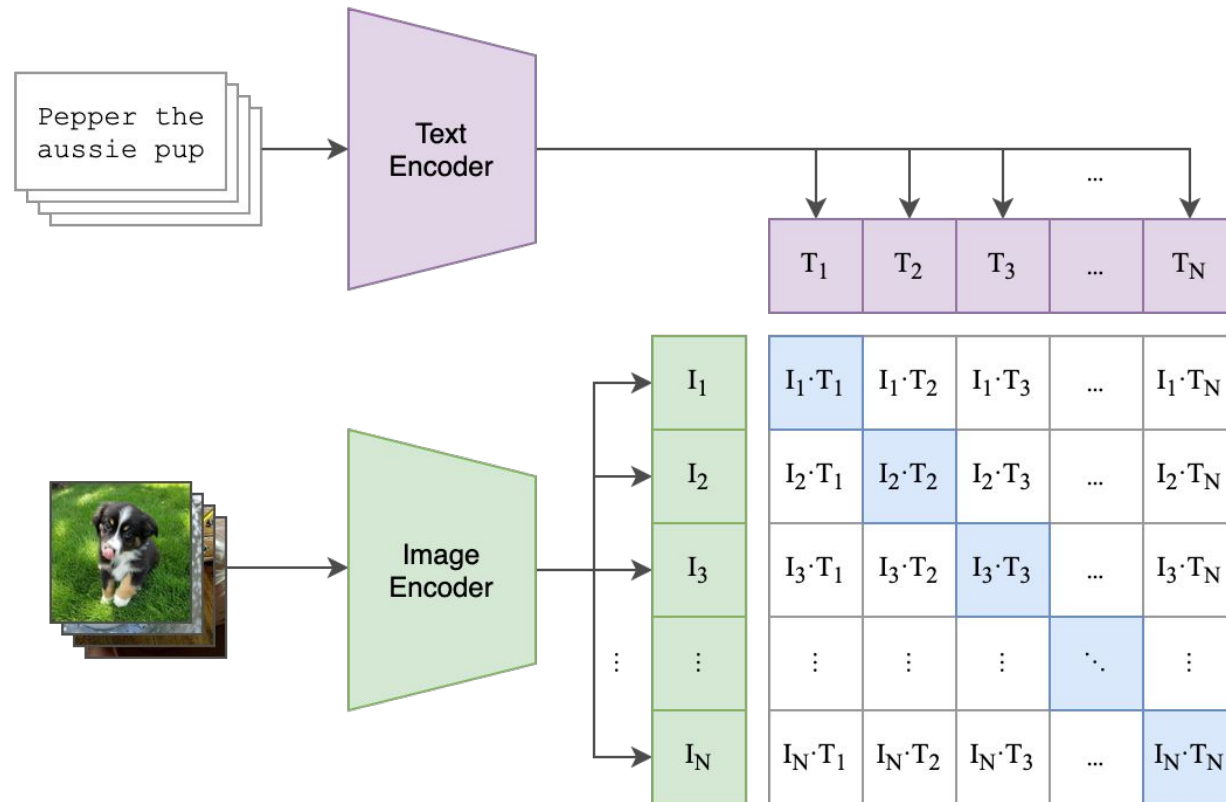
Decouples negative sample size from minibatch size; allows large batch training without TPUs.

<https://arxiv.org/abs/1911.05722>

Contrastive Language Image Pre-Training (CLIP)



(1) Contrastive pre-training



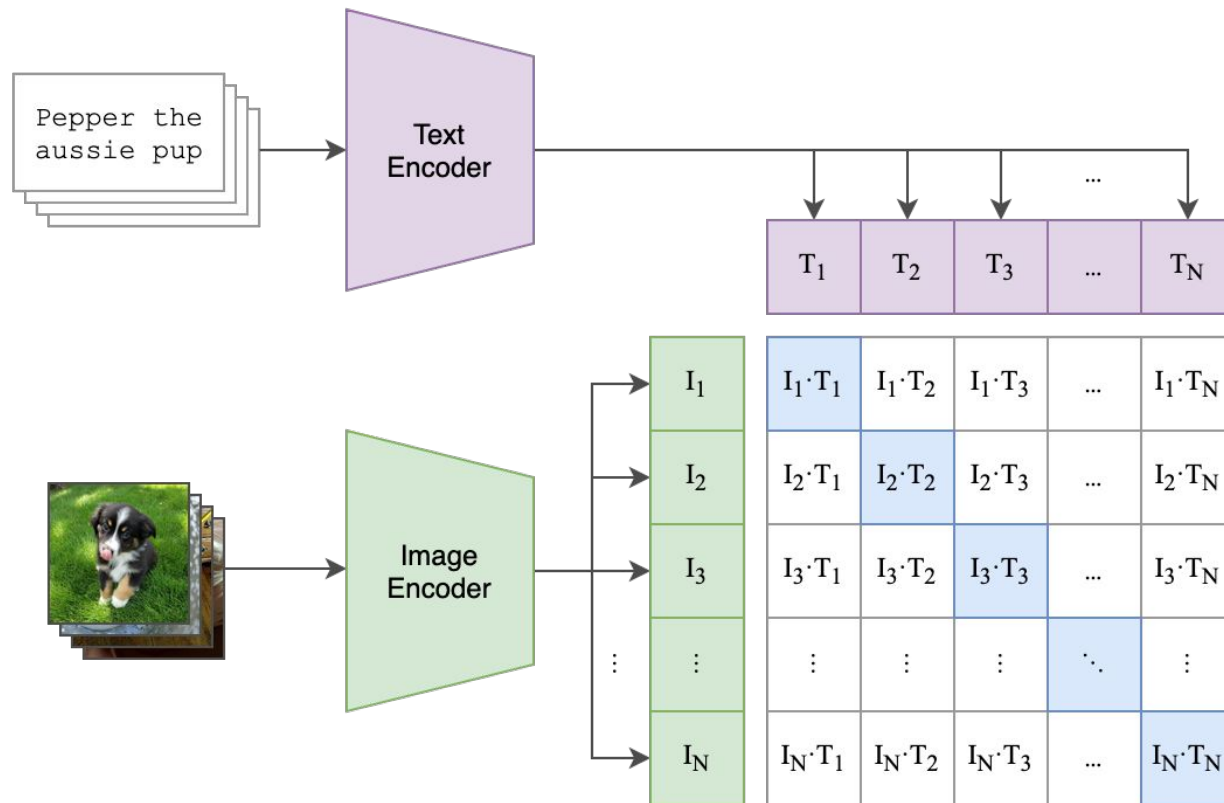
$$\text{InfoNCE loss} = -\log \frac{\exp I_i T_i}{\sum_j \exp I_i T_j}$$

Contrastive Language Image Pre-Training (CLIP)

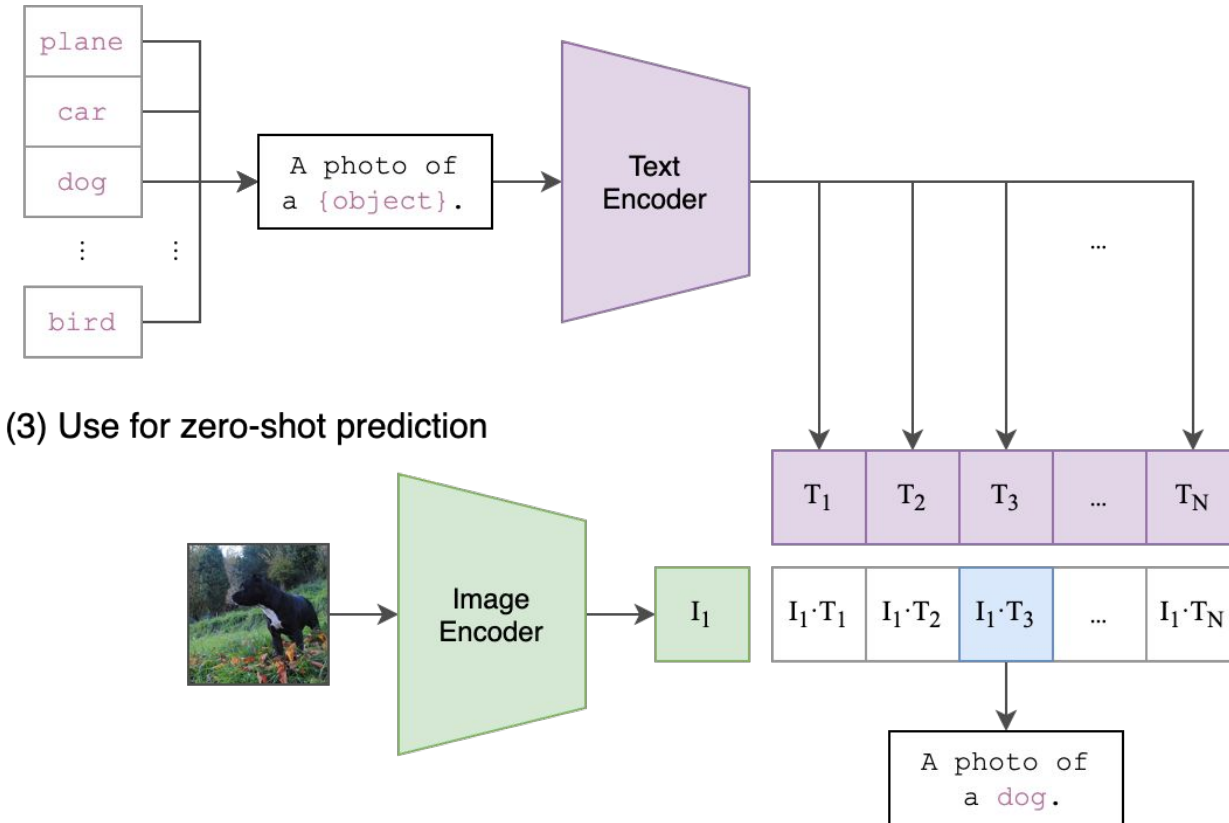


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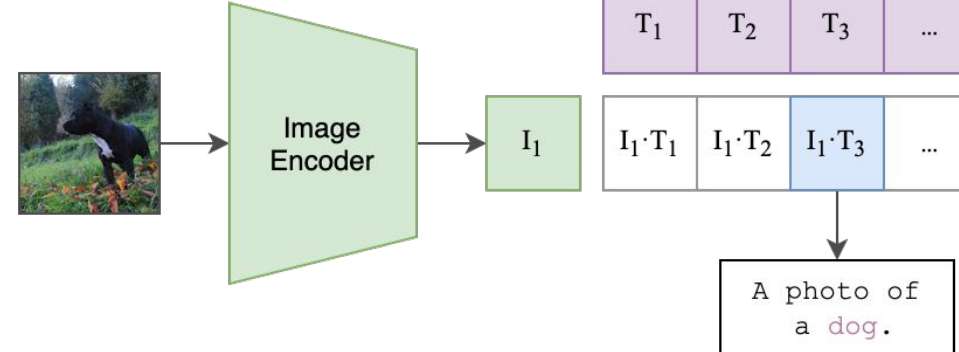
(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction





What is a key disadvantage of CLIP (and many other contrastive methods)?

Diffusion Models

- Discriminative
 - Rotation
 - Jigsaw
 - Clustering
 - Contrastive Learning
- Diffusion Models

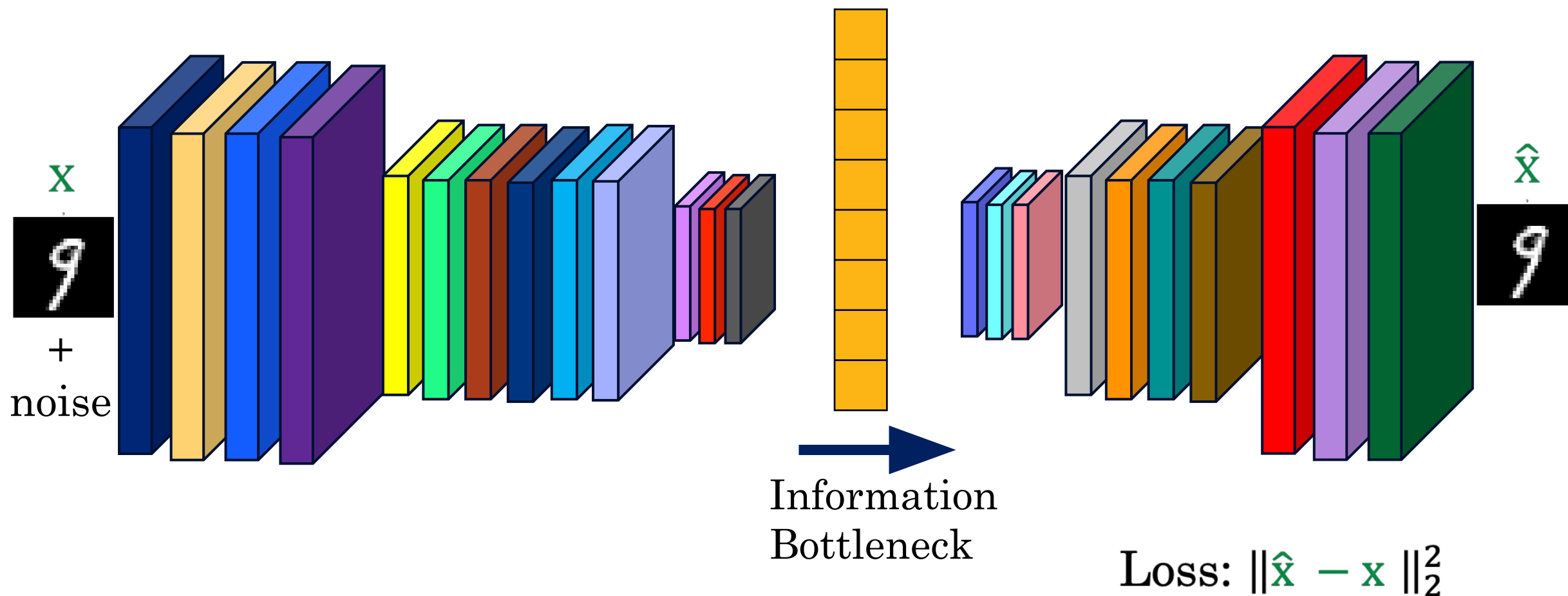


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Recall the Denoising Problem



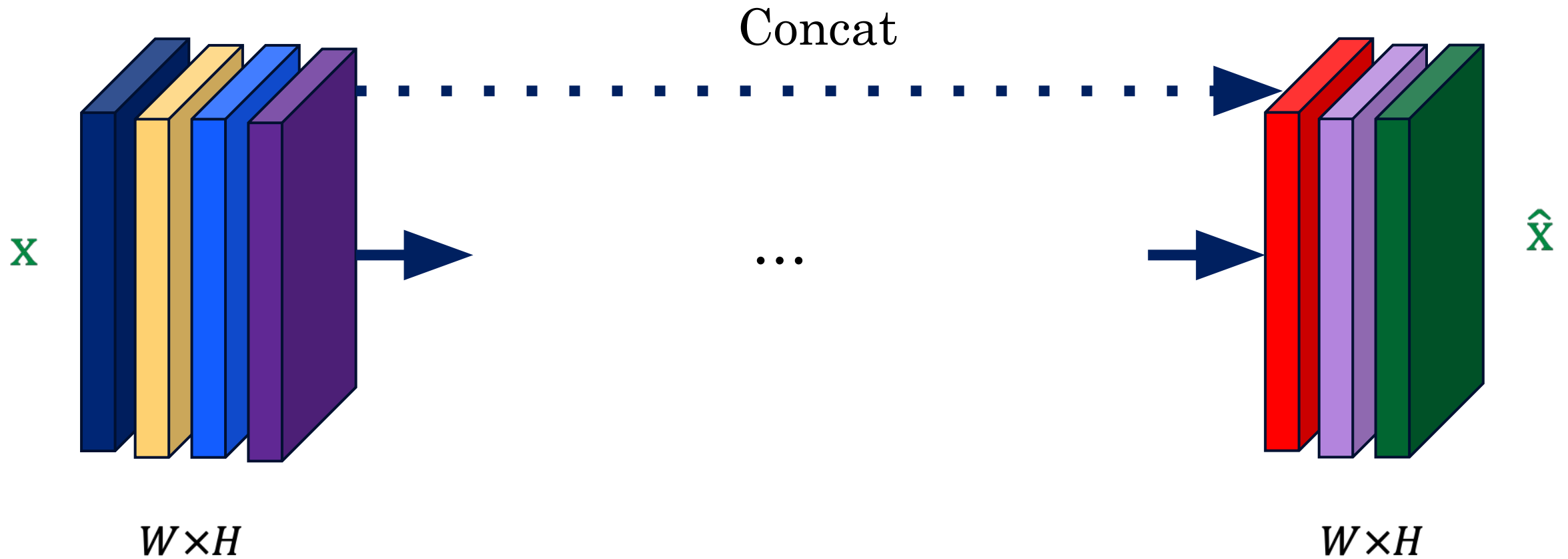
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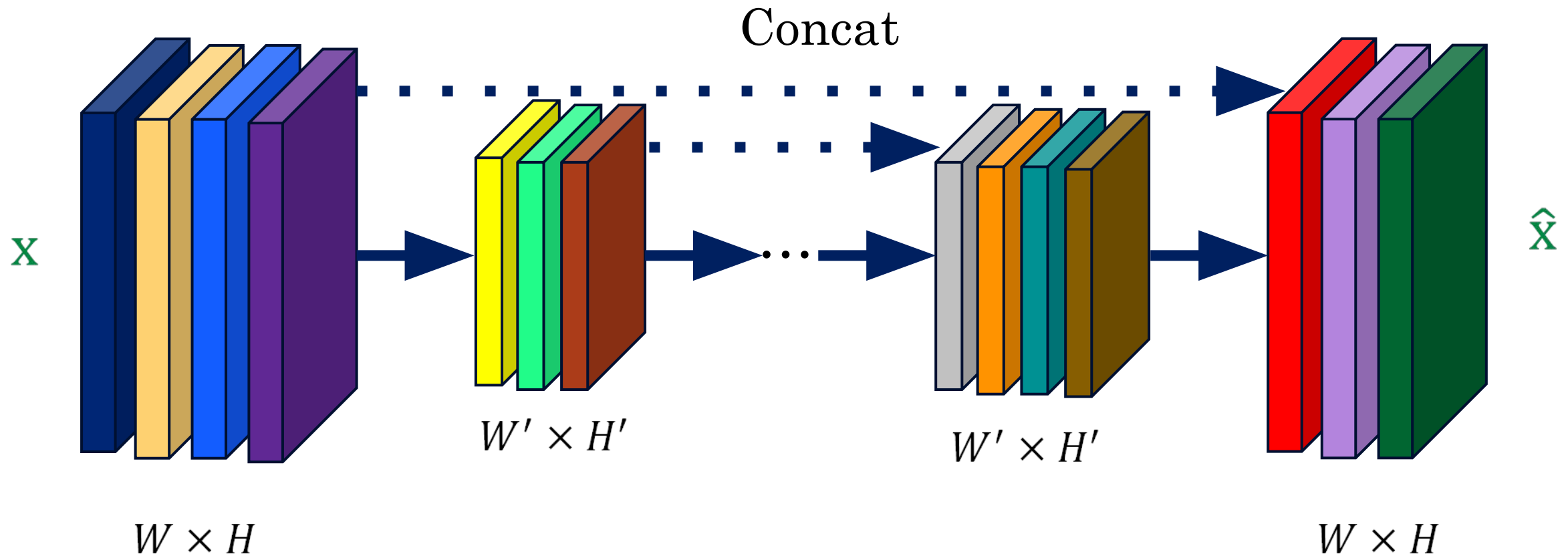


What are some shortcomings of traditional CNNs with an information bottleneck?

U-Net: Another Encoder-Decoder Architecture



U-Net: Another Encoder-Decoder Architecture



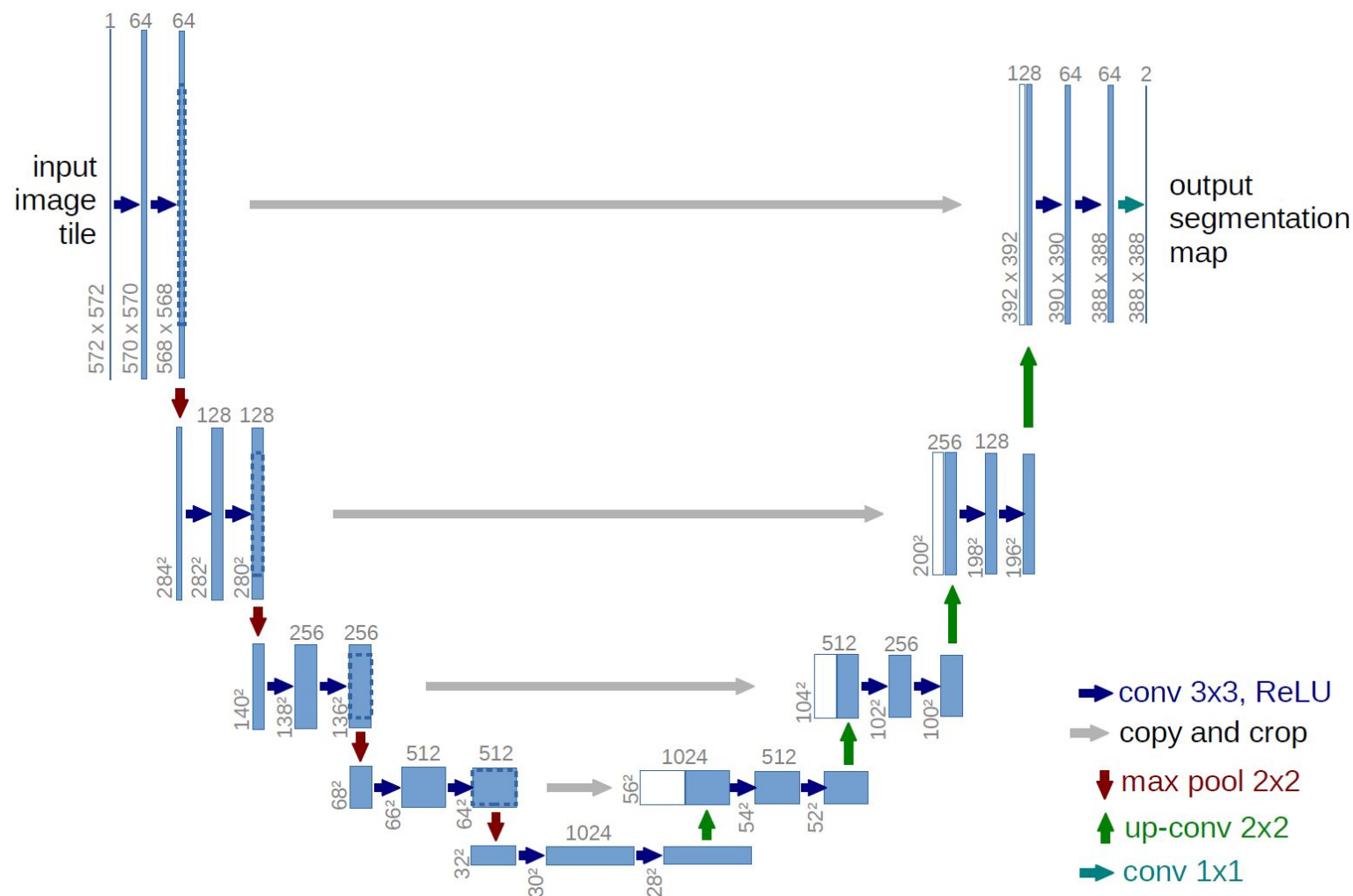
U-Net: Another Encoder-Decoder Architecture

<https://arxiv.org/pdf/1505.04597>

Published in MICCAI (Medical Image Computing and Computer Assisted Interventions) 2015



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Back to Denoising

From Noise to an Image



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From Noise to an Image



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From Noise to an Image



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From Noise to an Image



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From Noise to an Image



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From Noise to an Image



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From Noise to an Image



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From Noise to an Image



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From Noise to an Image



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From Noise to an Image



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From Noise to an Image



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From Noise to an Image



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From Noise to an Image



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From Noise to an Image



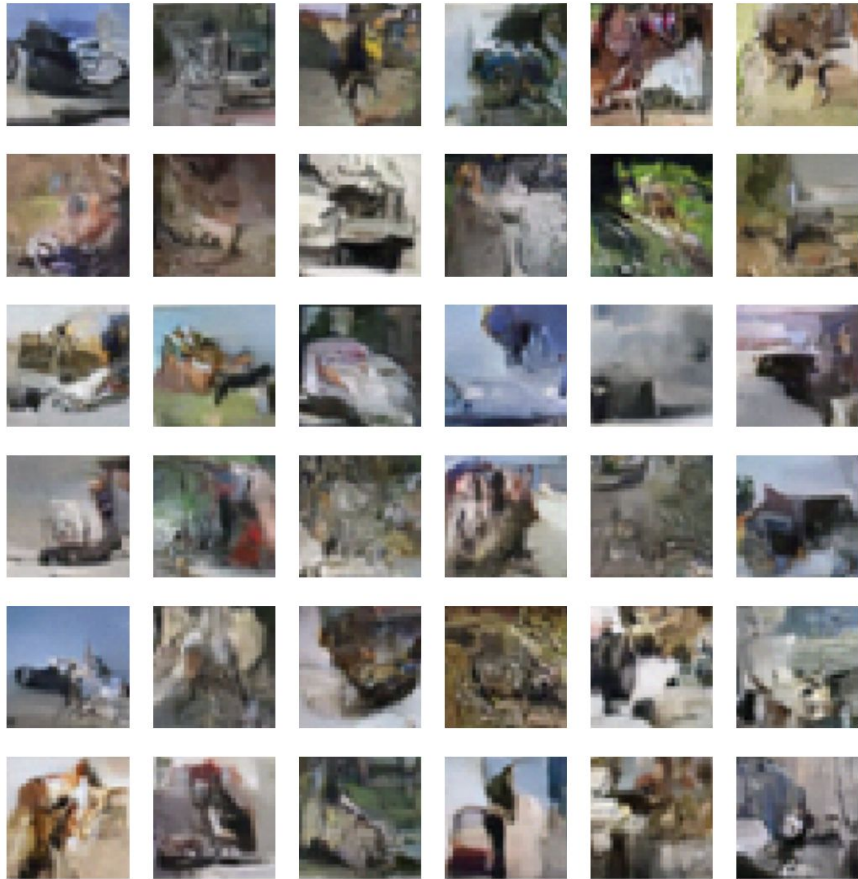
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From Noise to an Image



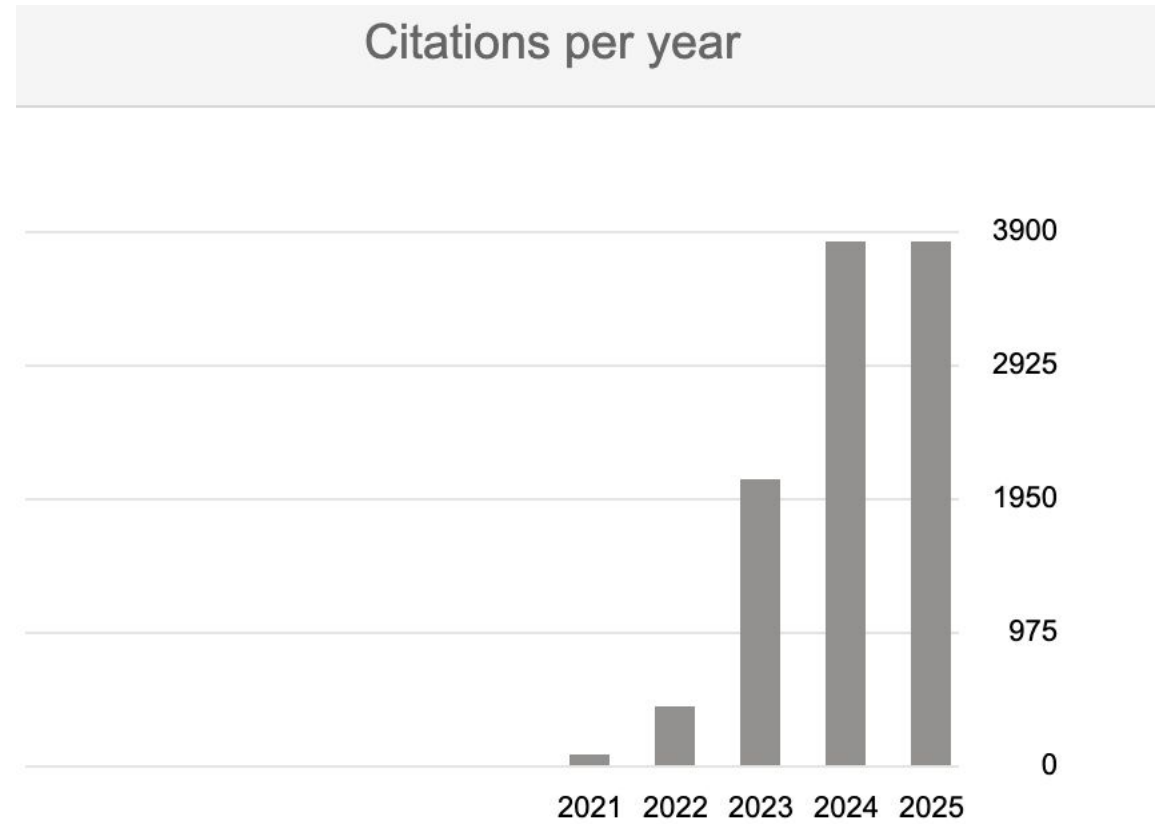
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Diffusion Models



[Sohl-Dickstein et al., 2015]

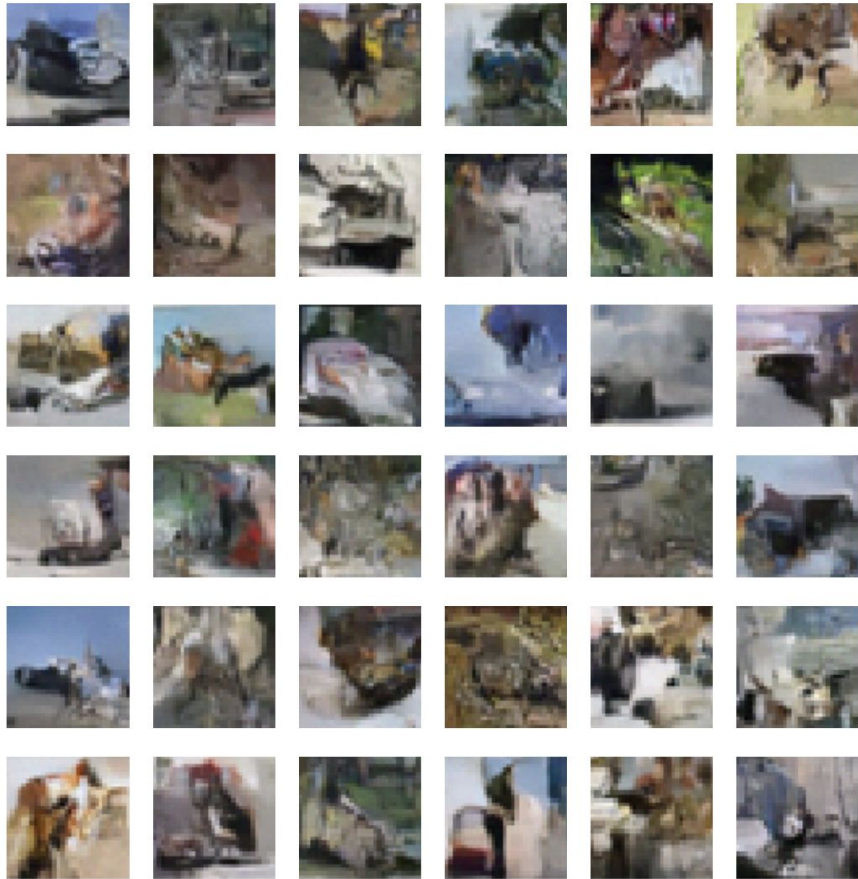
Deep unsupervised learning using nonequilibrium thermodynamics



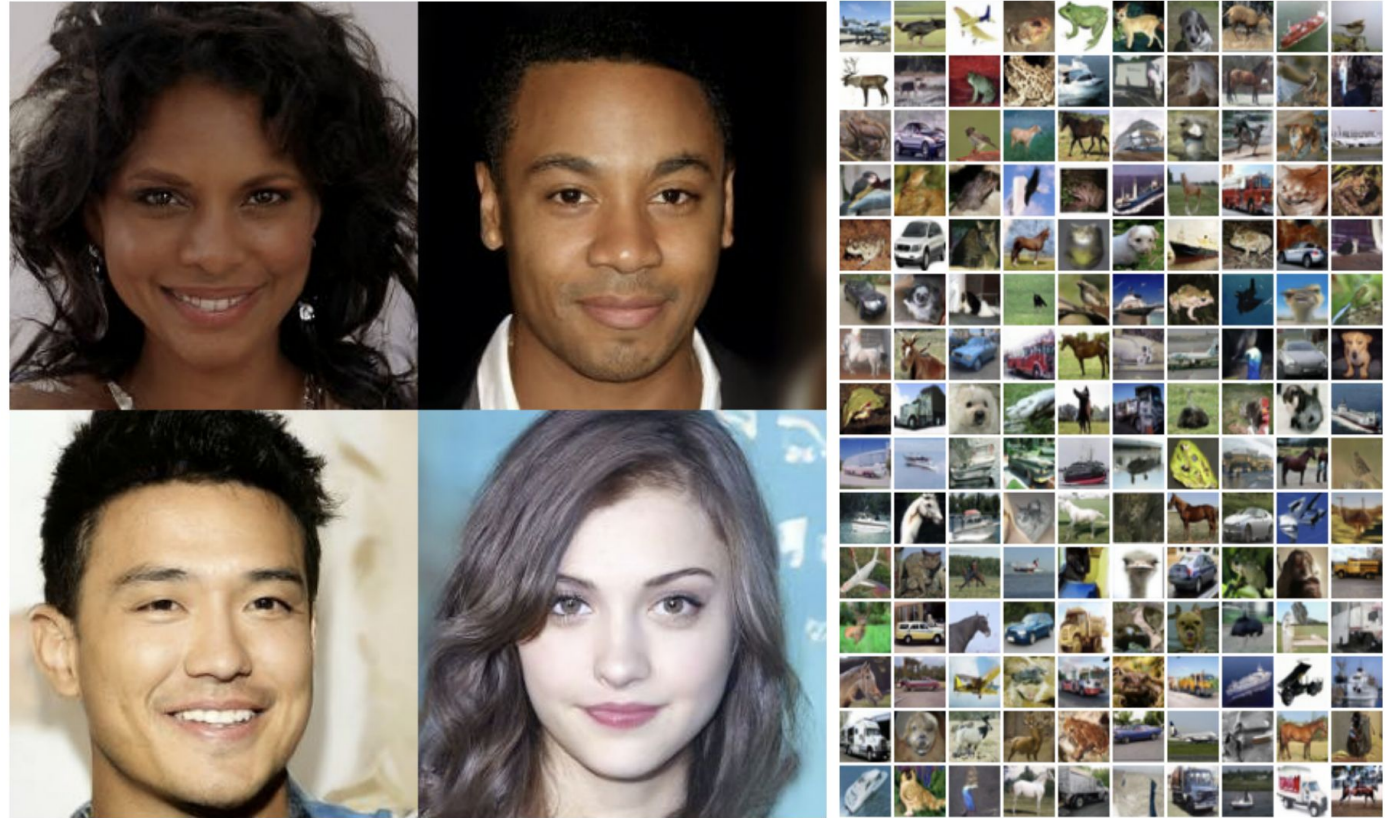
Diffusion Models



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[Sohl-Dickstein et al., 2015]

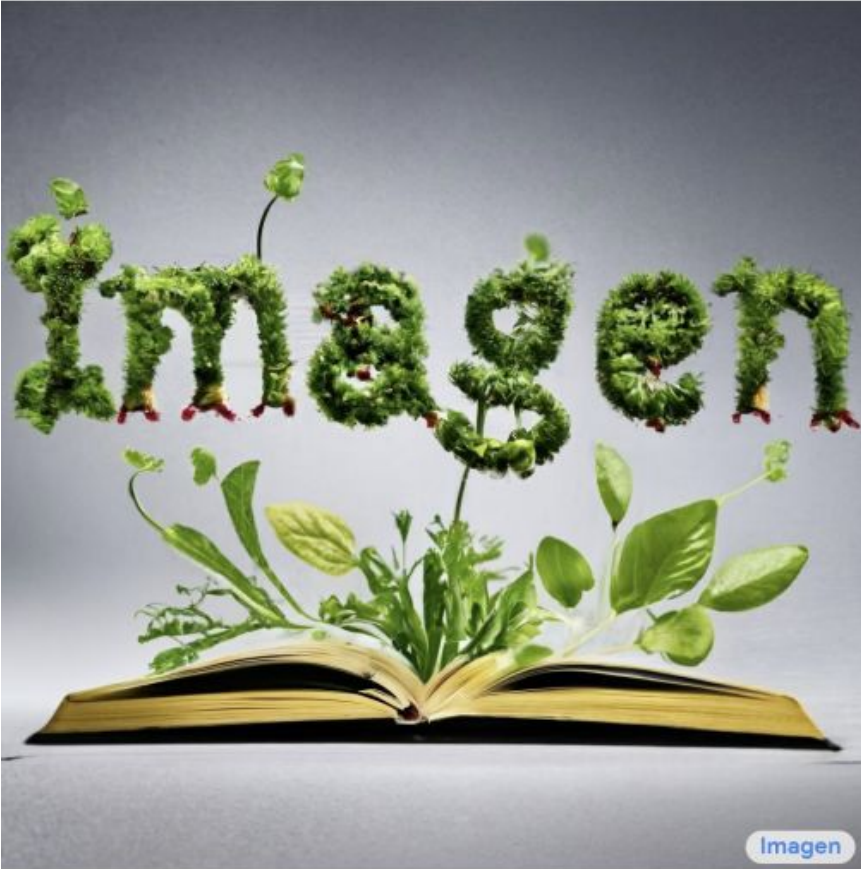


[Ho, et al., “Denoising Diffusion”, 2020]

Imagen



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Sprouts in the shape of text 'Imagen' coming out of a fairytale book.



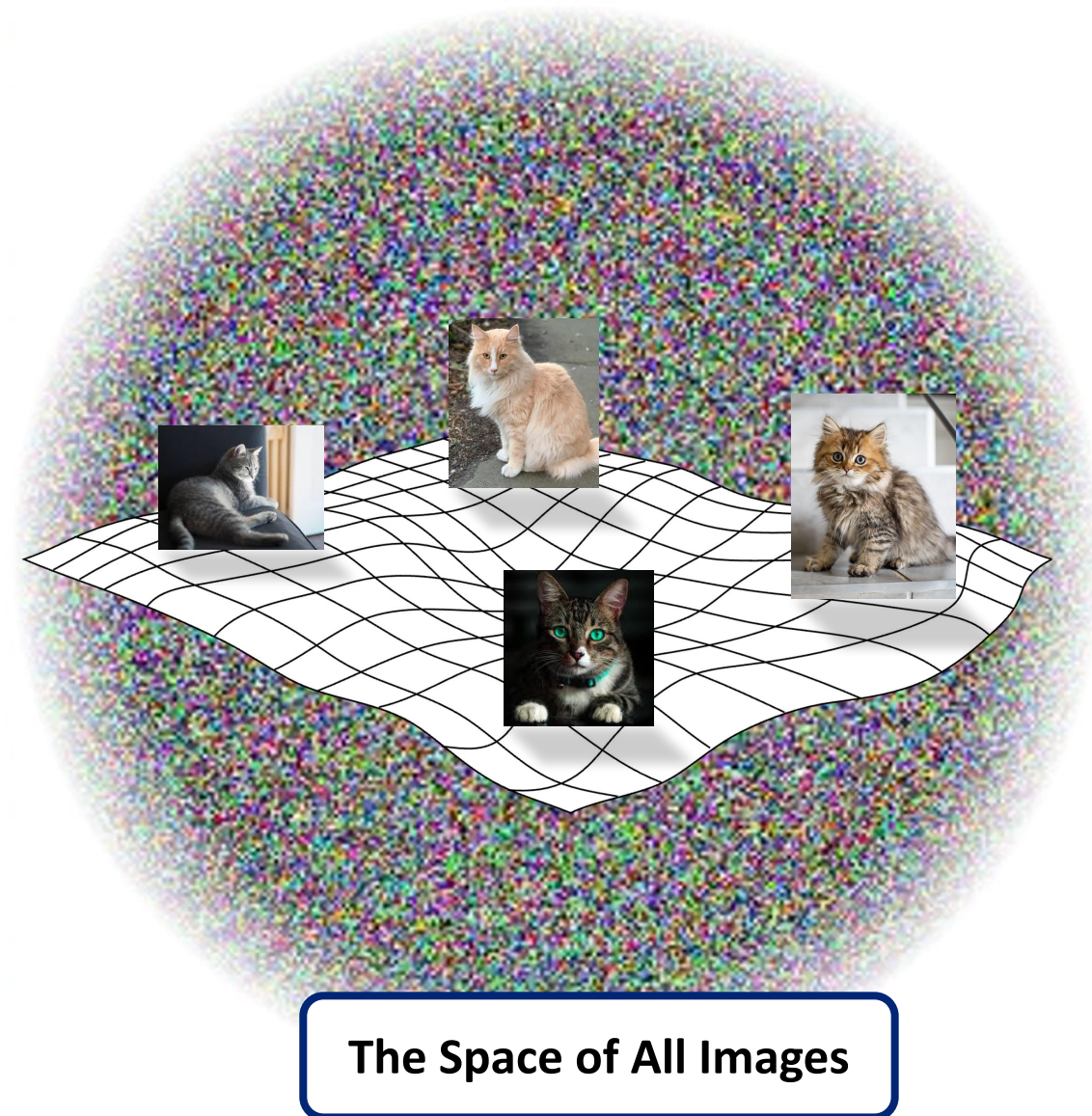
A photo of a Shiba Inu dog with a backpack riding a bike. It is wearing sunglasses and a beach hat.

Recall: Natural Image Manifolds



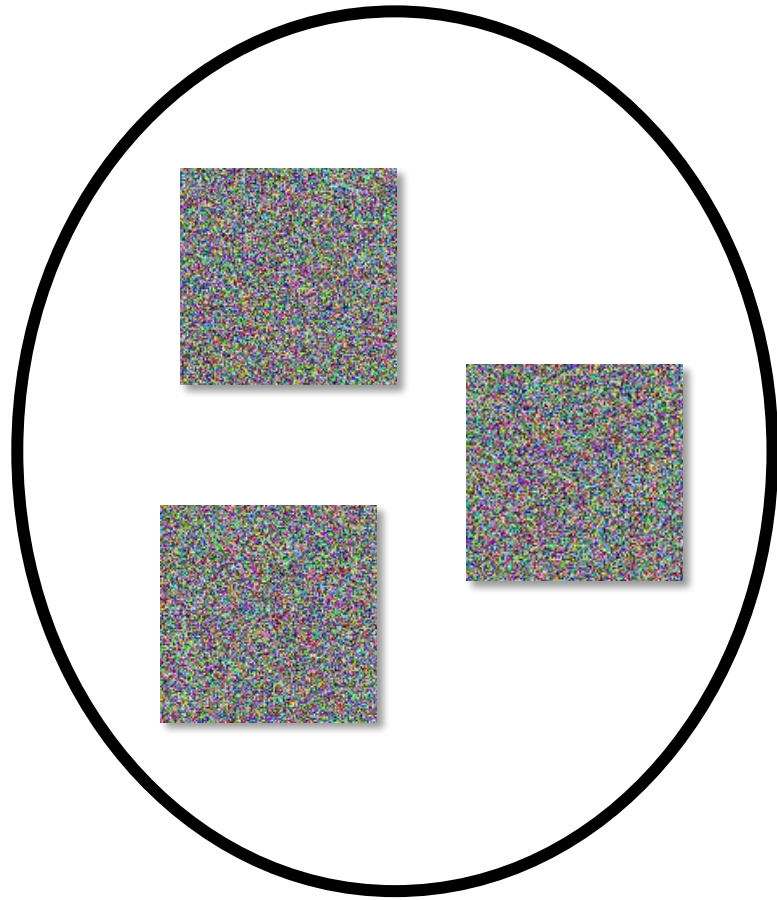
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- Most images are “noise”.
- “Meaningful” images tend to form some manifold within the space of all images.
- Images of a particular class fall on manifolds within that manifold...



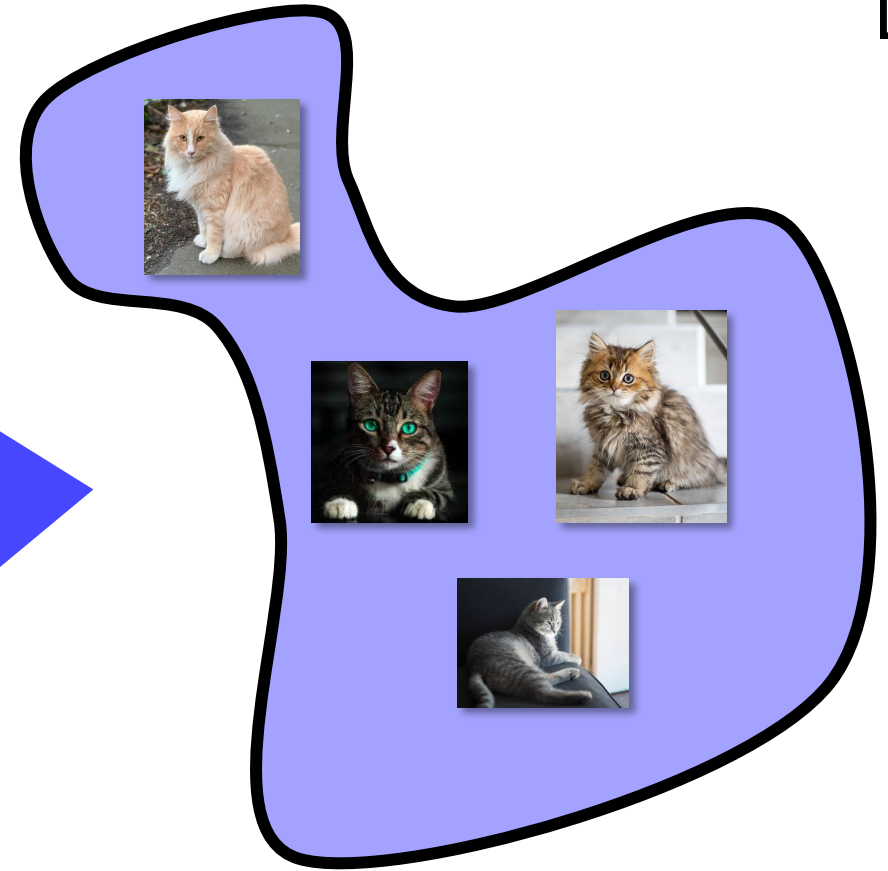


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Random
images

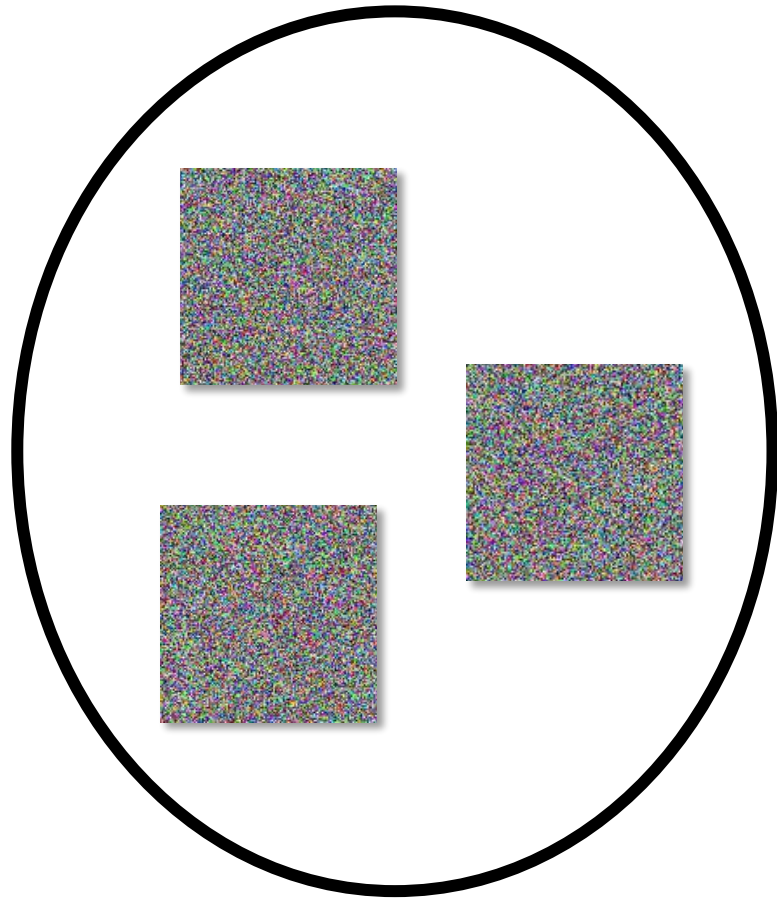
Diffusion



Manifold of cat
images

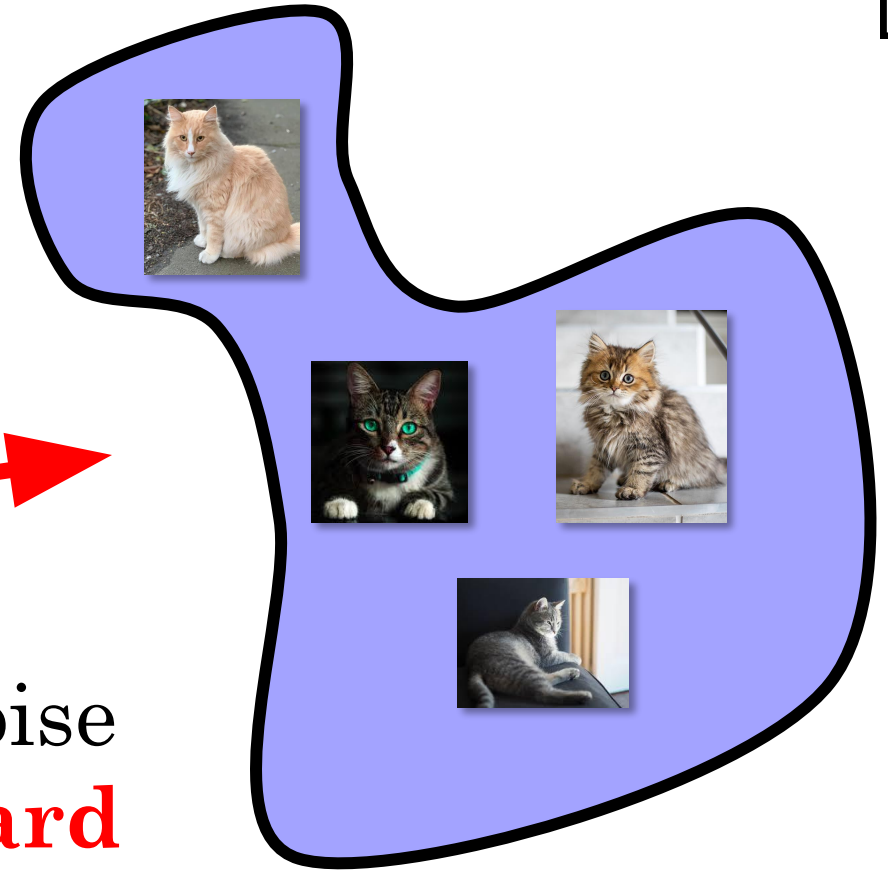


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Random
images

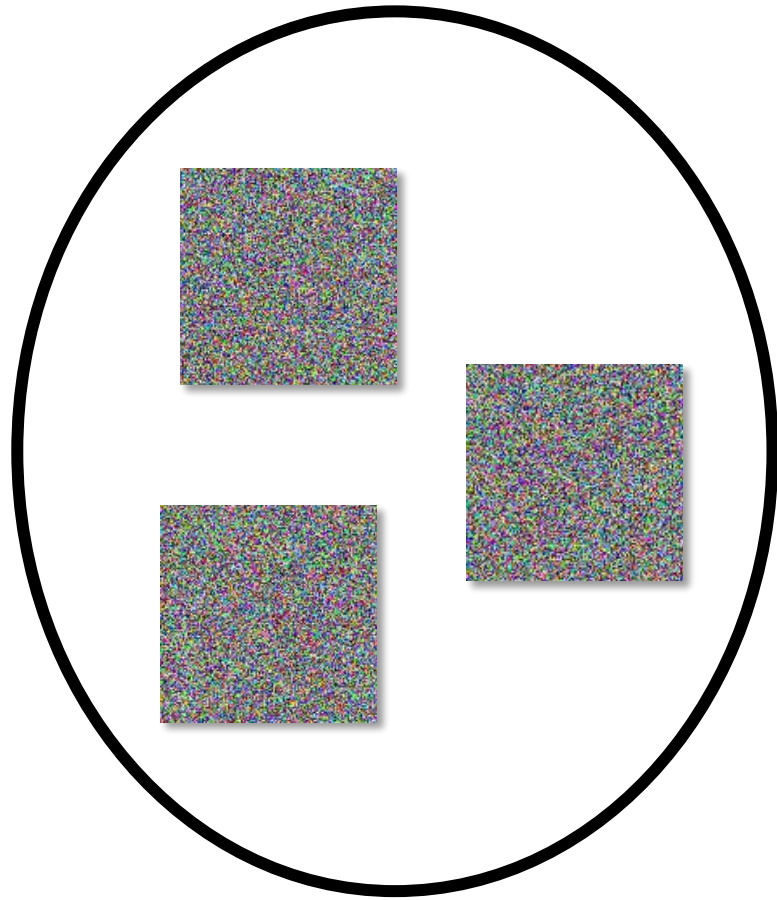
Forward
mapping (noise
to cats) is **hard**



Manifold of cat
images

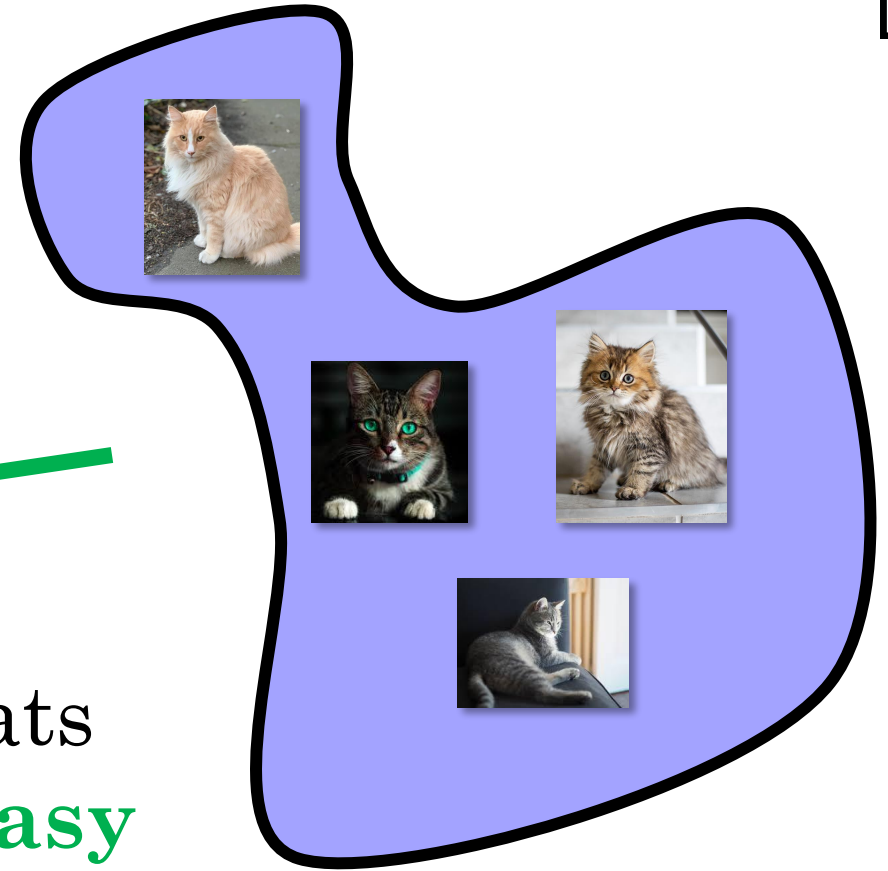


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Random
images

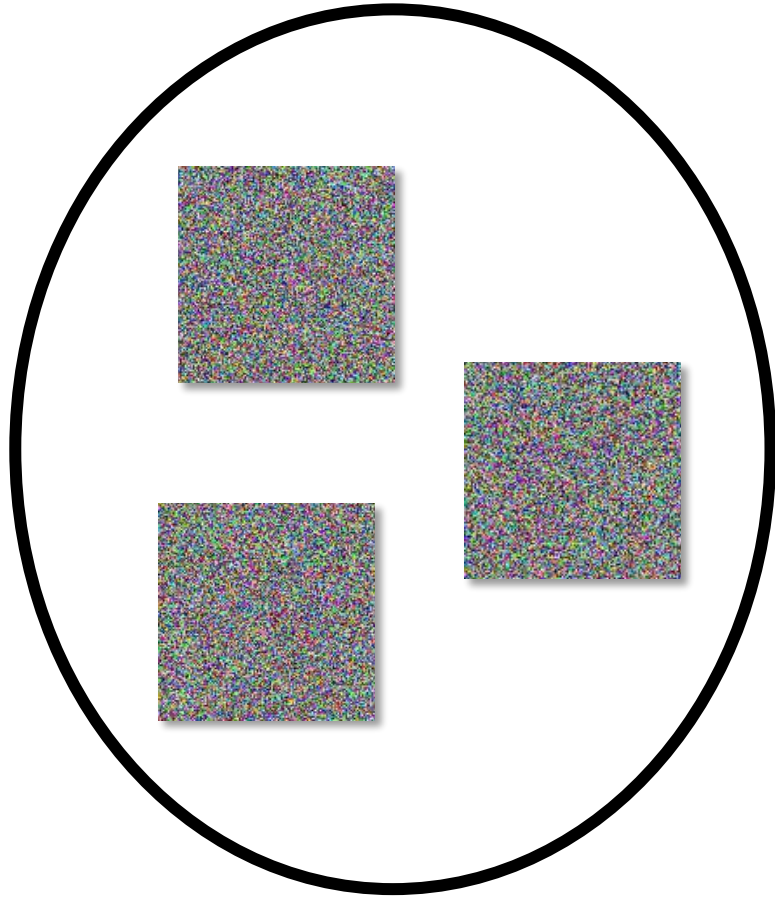
Reverse
mapping (cats
to noise) is **easy**



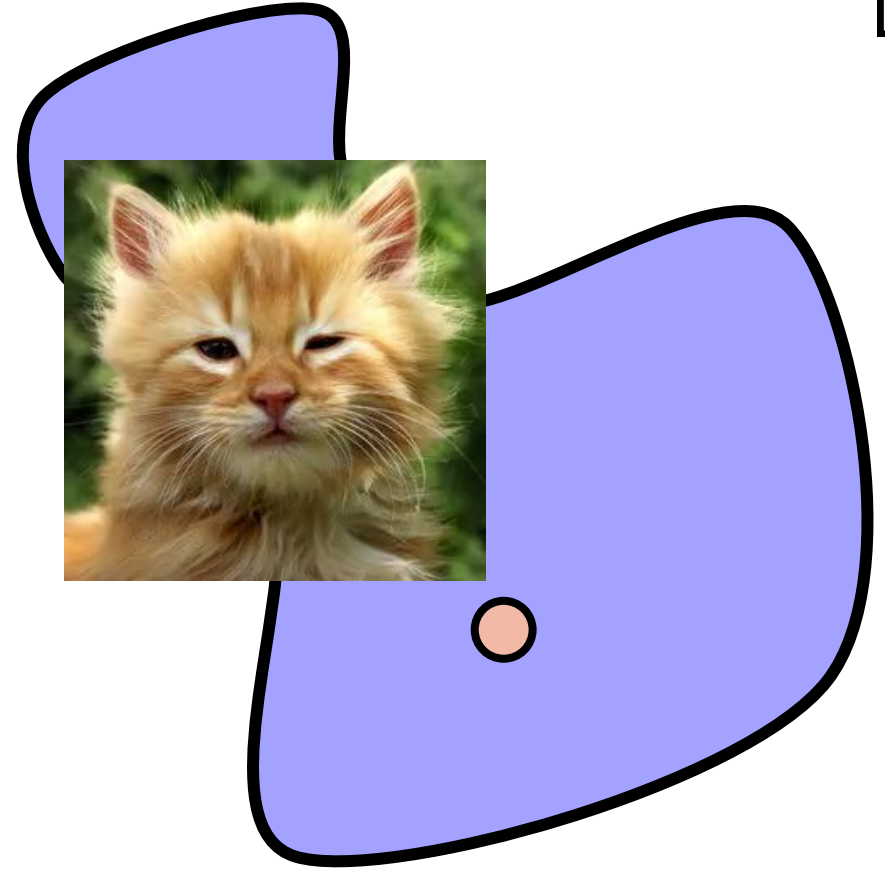
Manifold of cat
images



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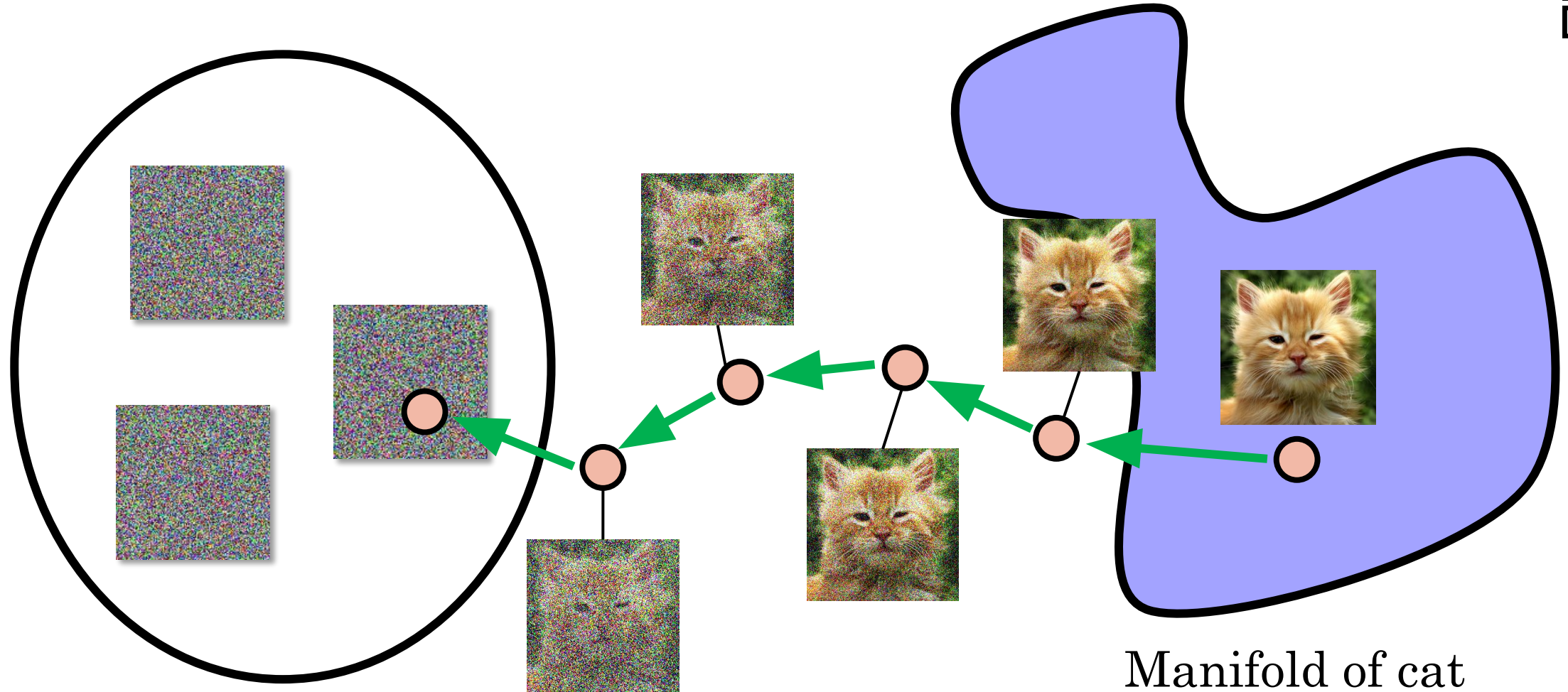
Random
images



Manifold of cat
images



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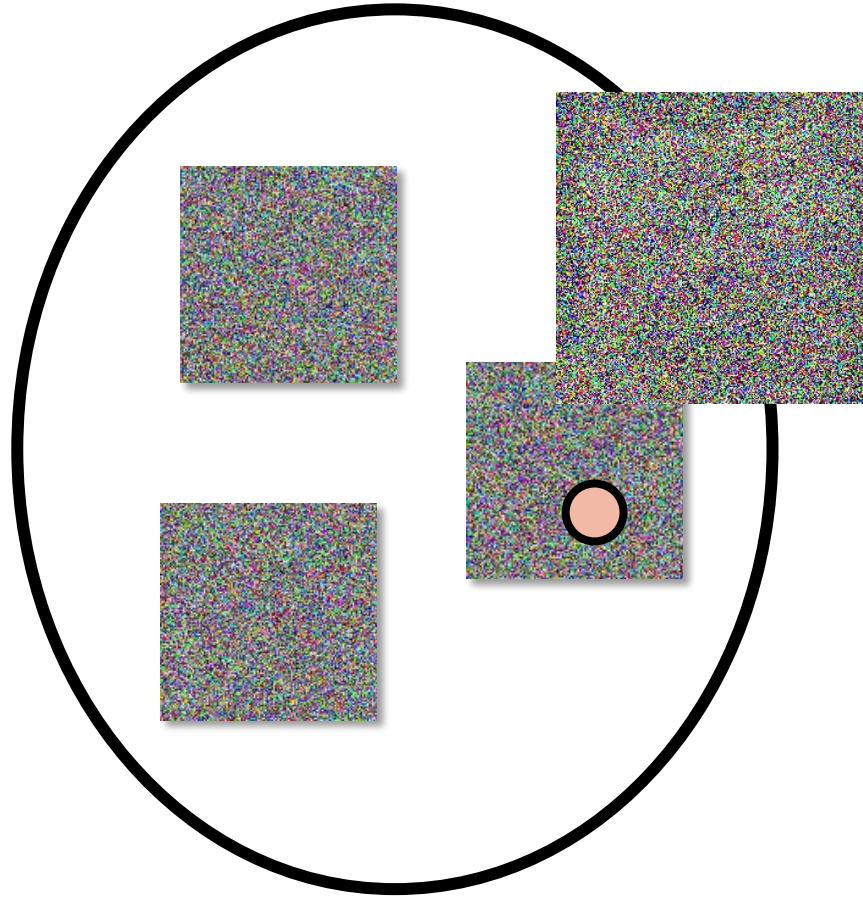


Random
images

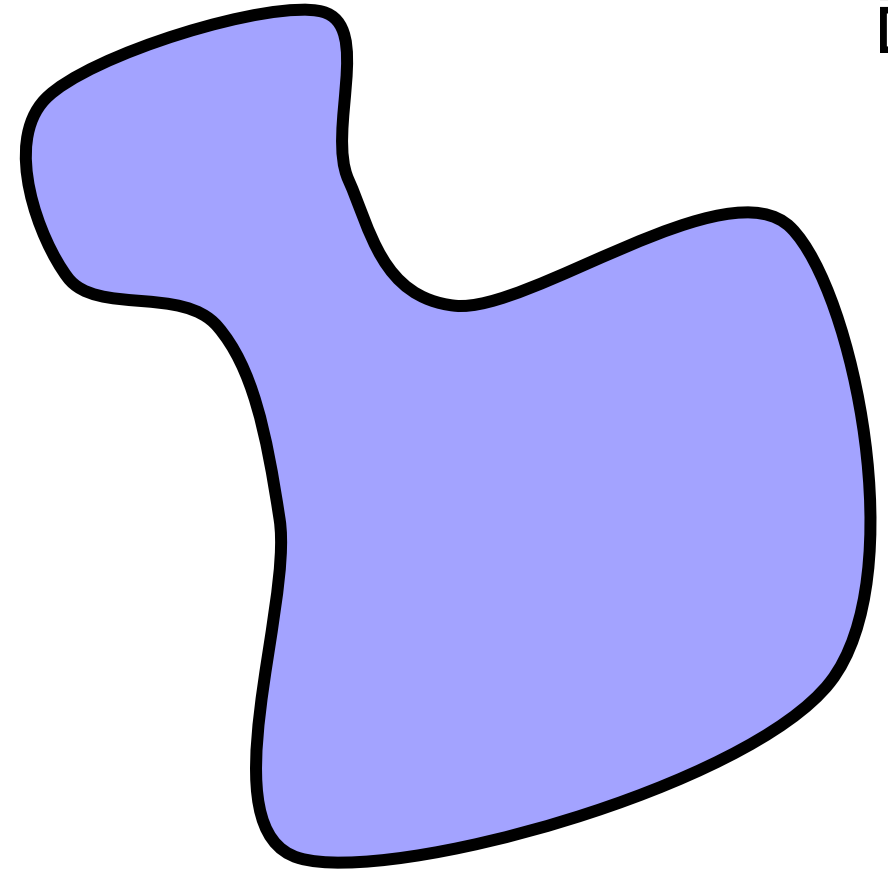
Manifold of cat
images



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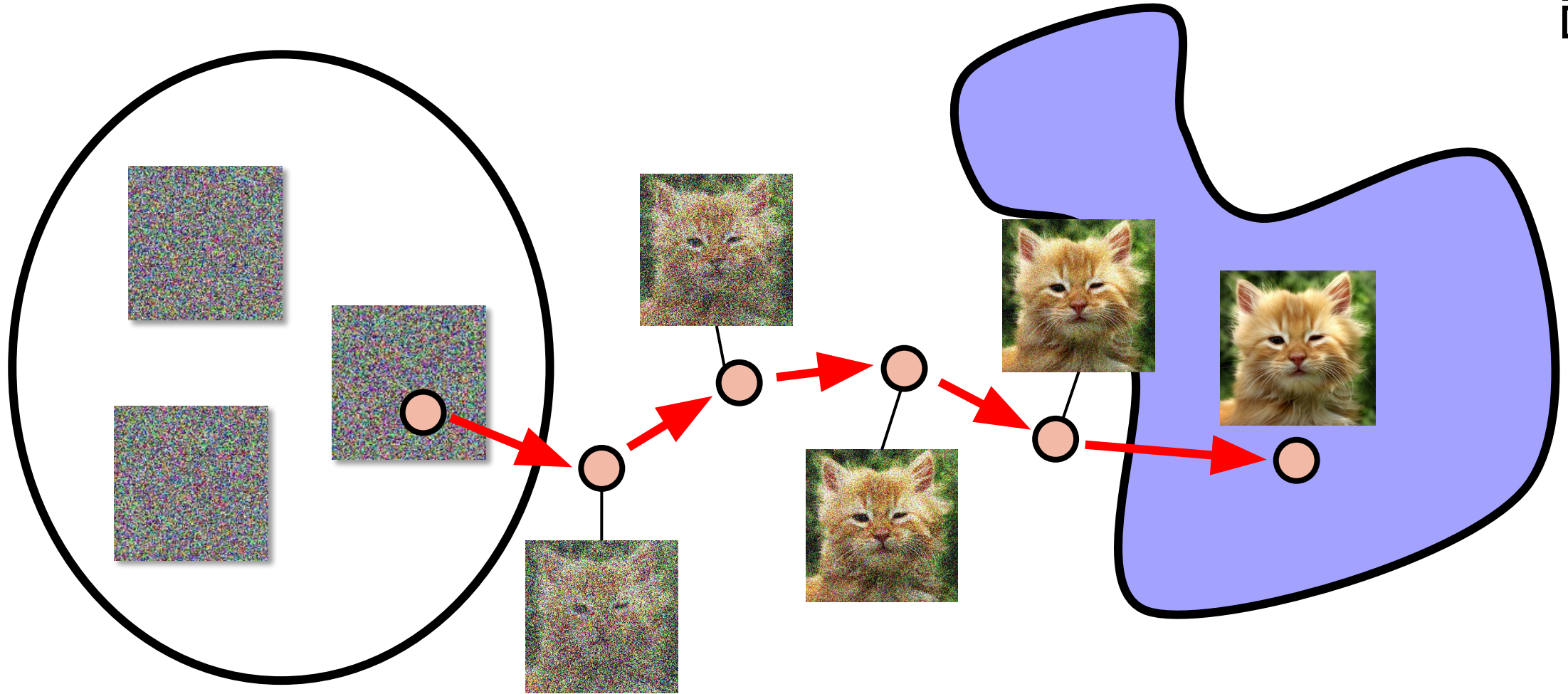
Random
images



Manifold of cat
images



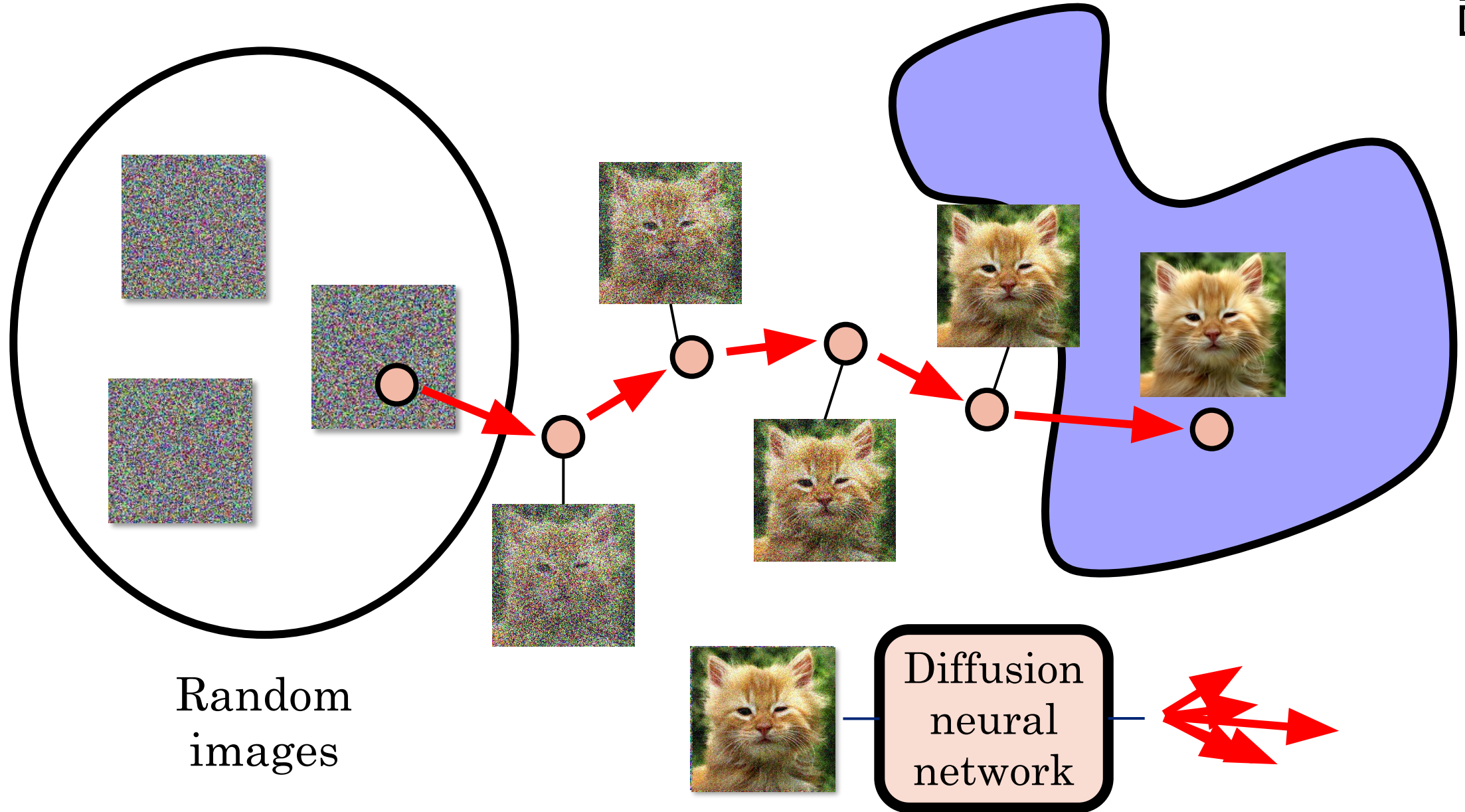
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Random
images

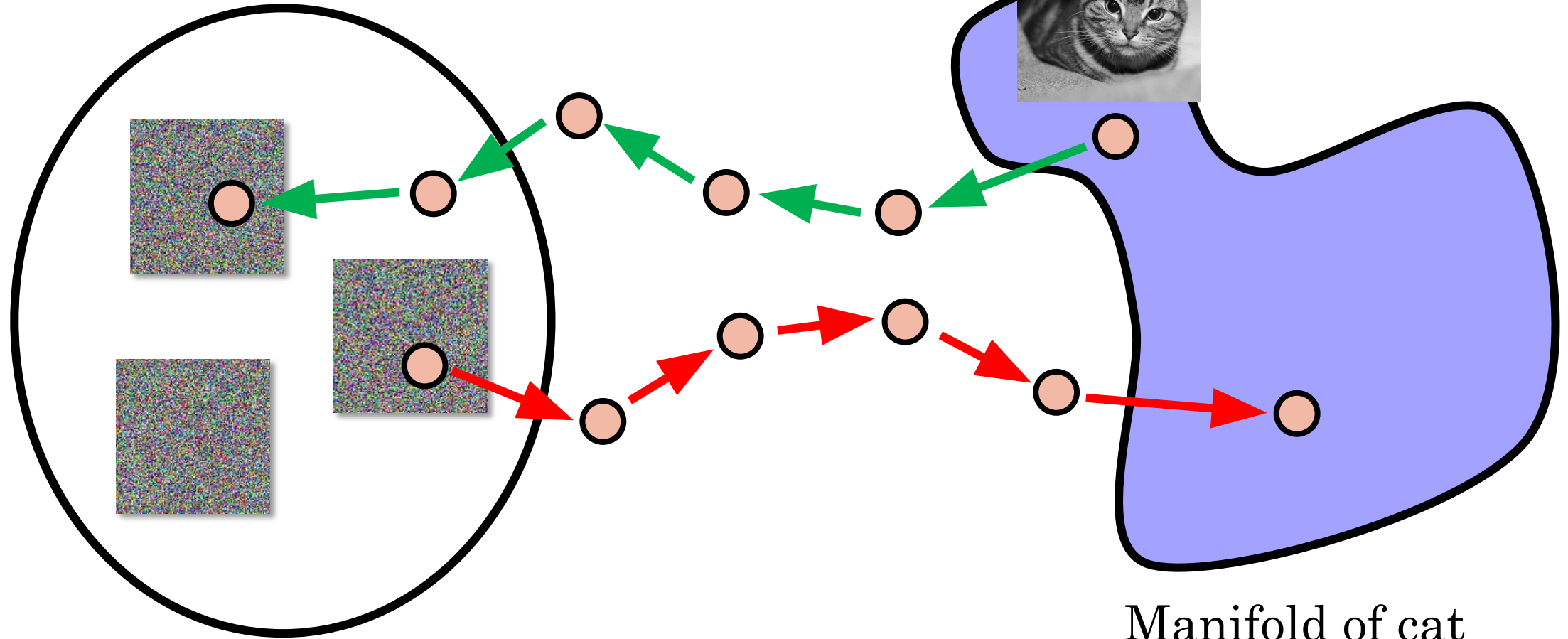


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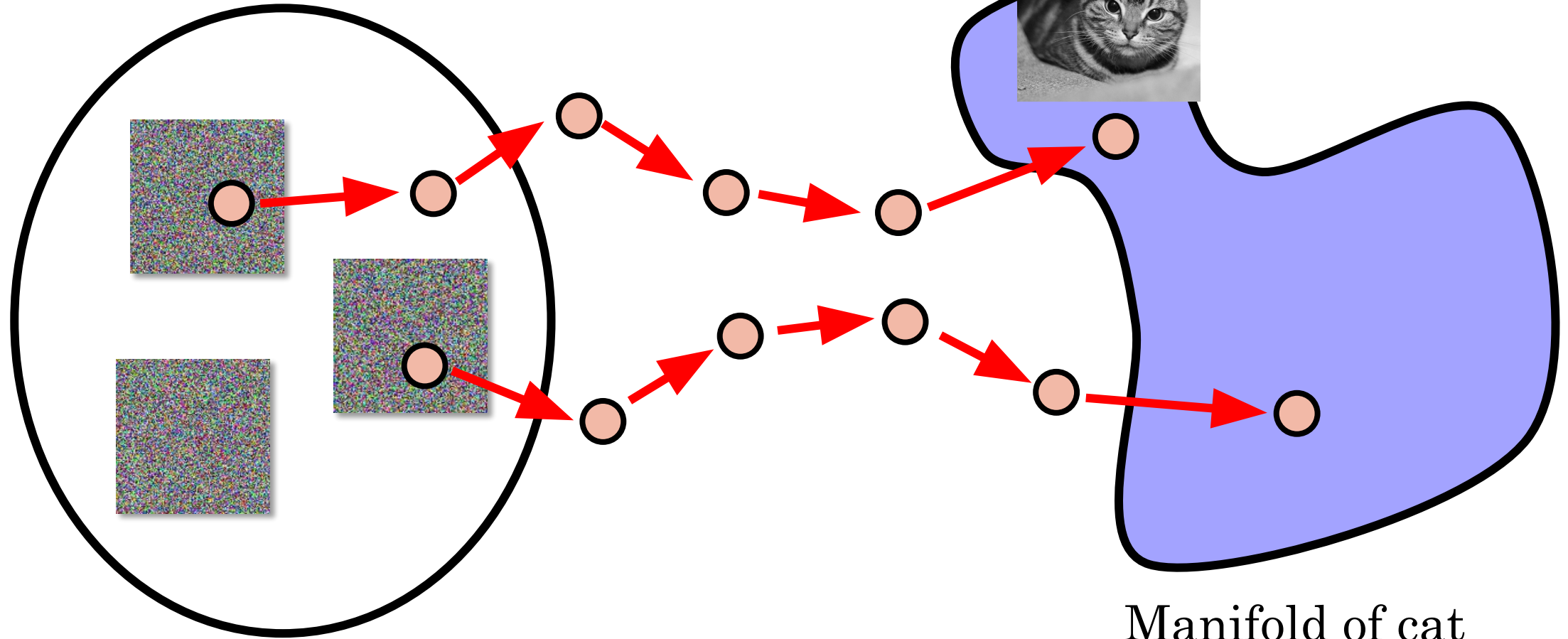


Random
images

Manifold of cat
images



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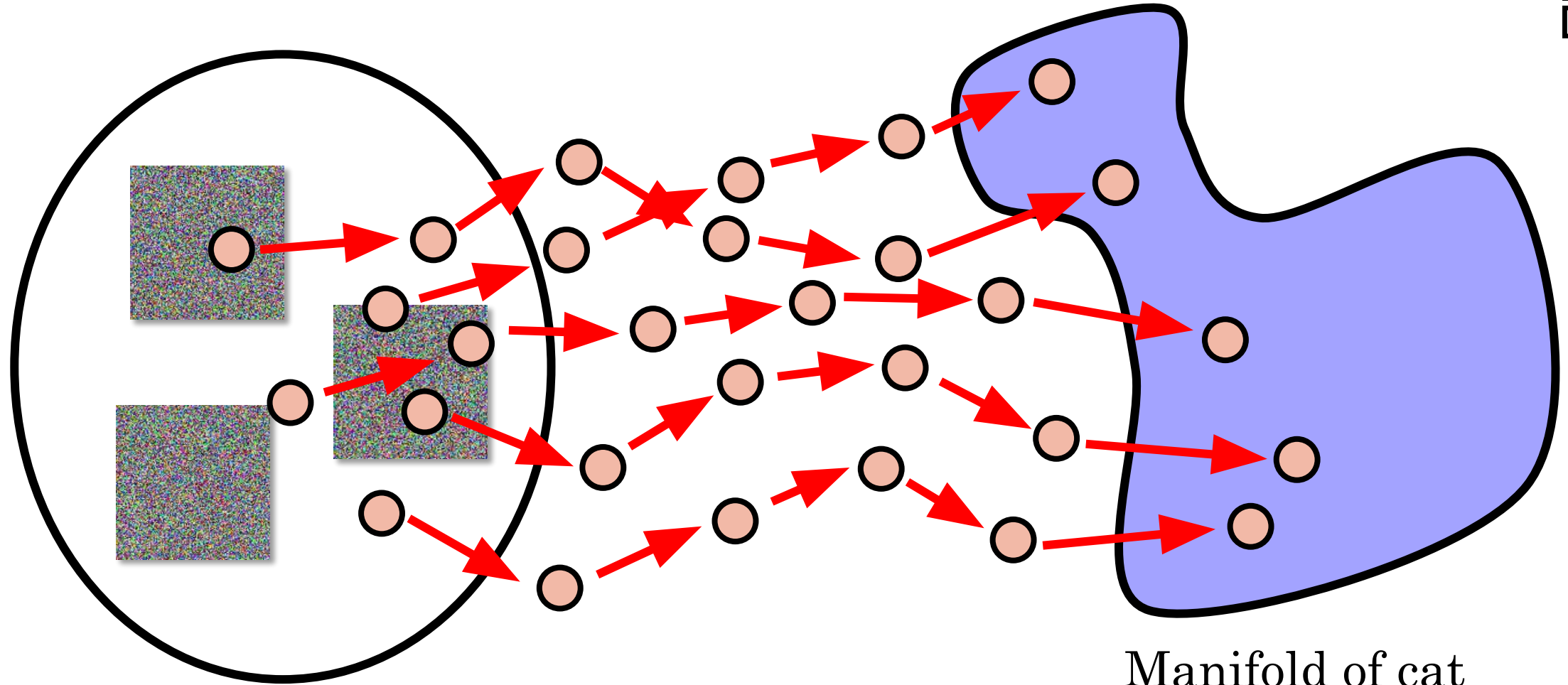


Random
images

Manifold of cat
images



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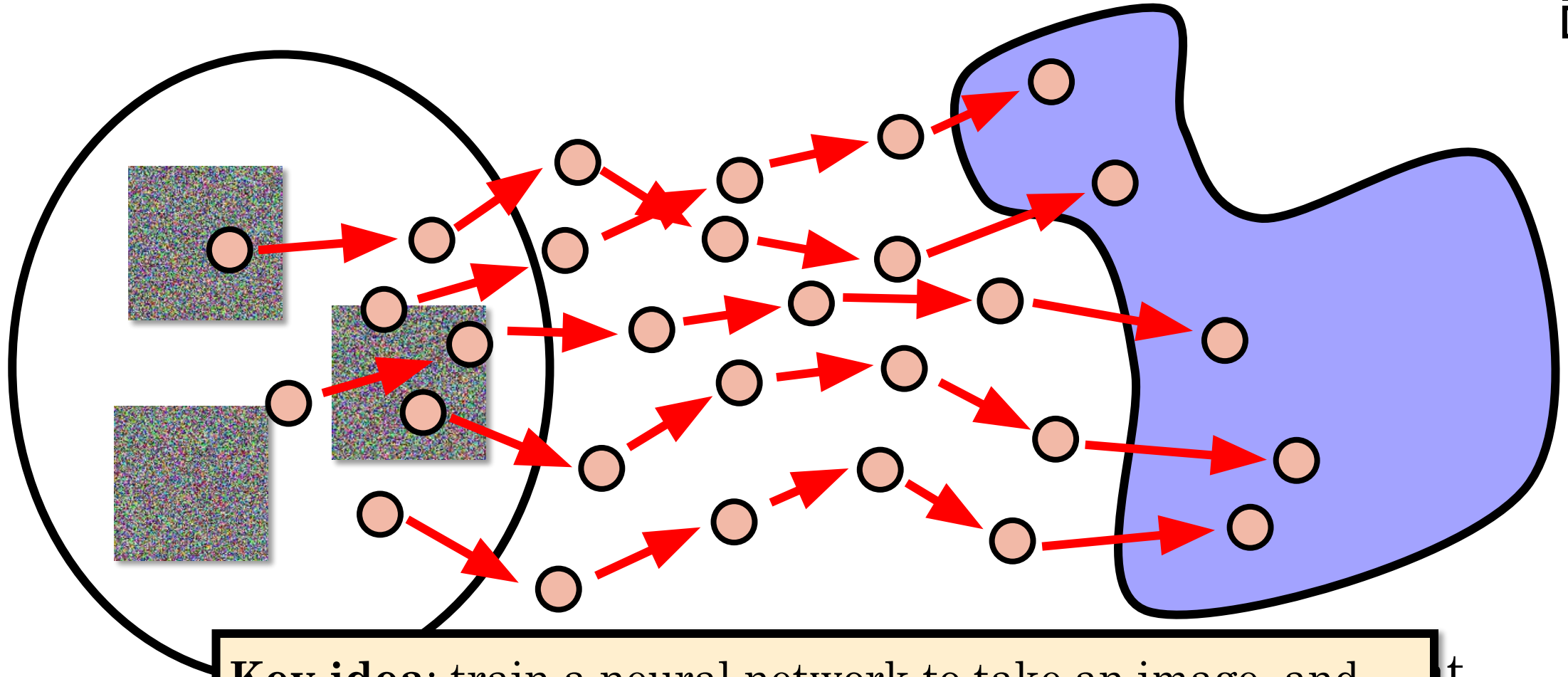


Random
images

Manifold of cat
images

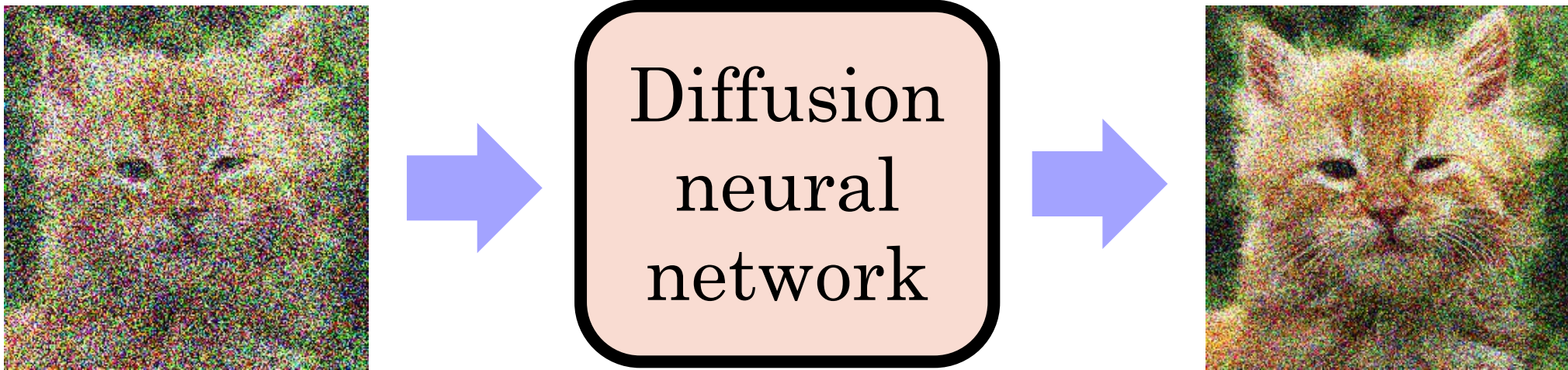


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Key idea: train a neural network to take an image, and predict the arrows above; that is, predict to convert a noisy image to a slightly less noisy image that is closer to the desired image manifold, using the example above to train.

Denoising Diffusion Neural Network

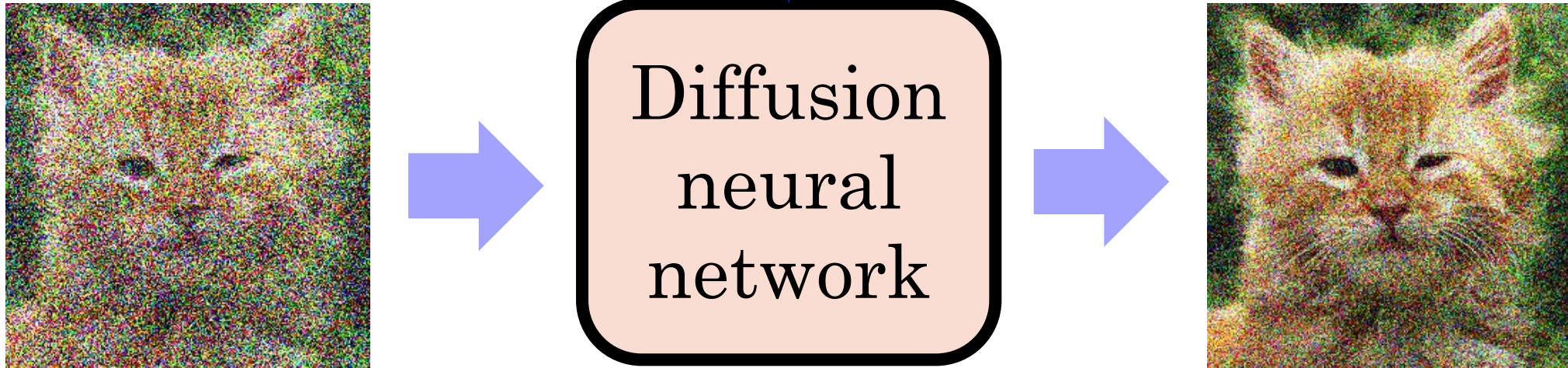


↑
This network can be a U-Net or
other image-to-image network.

Denoising Diffusion Neural Network



Timestep t (from say 0 to 999) is an additional input to the network (positionally encoded).

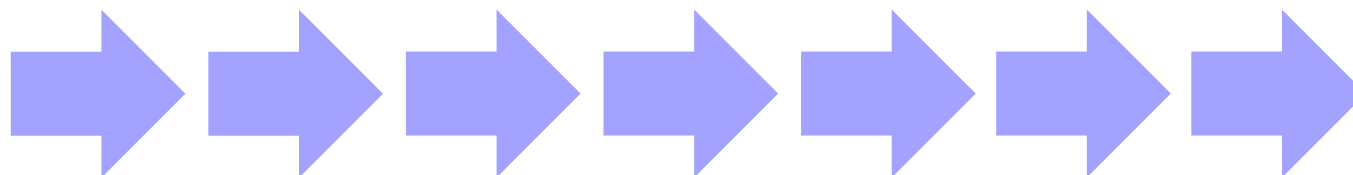


A basic diffusion approach will run this denoising for many timesteps (e.g., $T = 1000$ steps)

Sampling Many Outputs



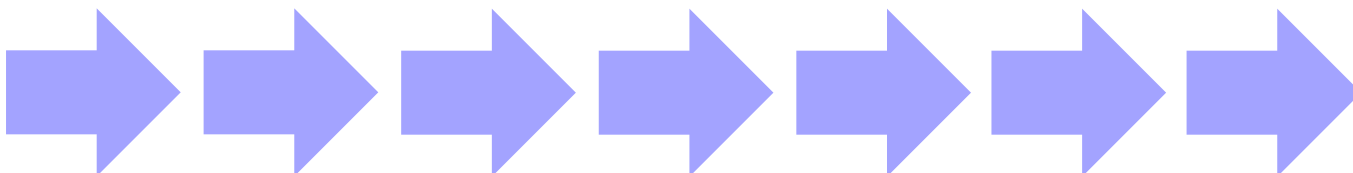
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Noise 1

Output image 1

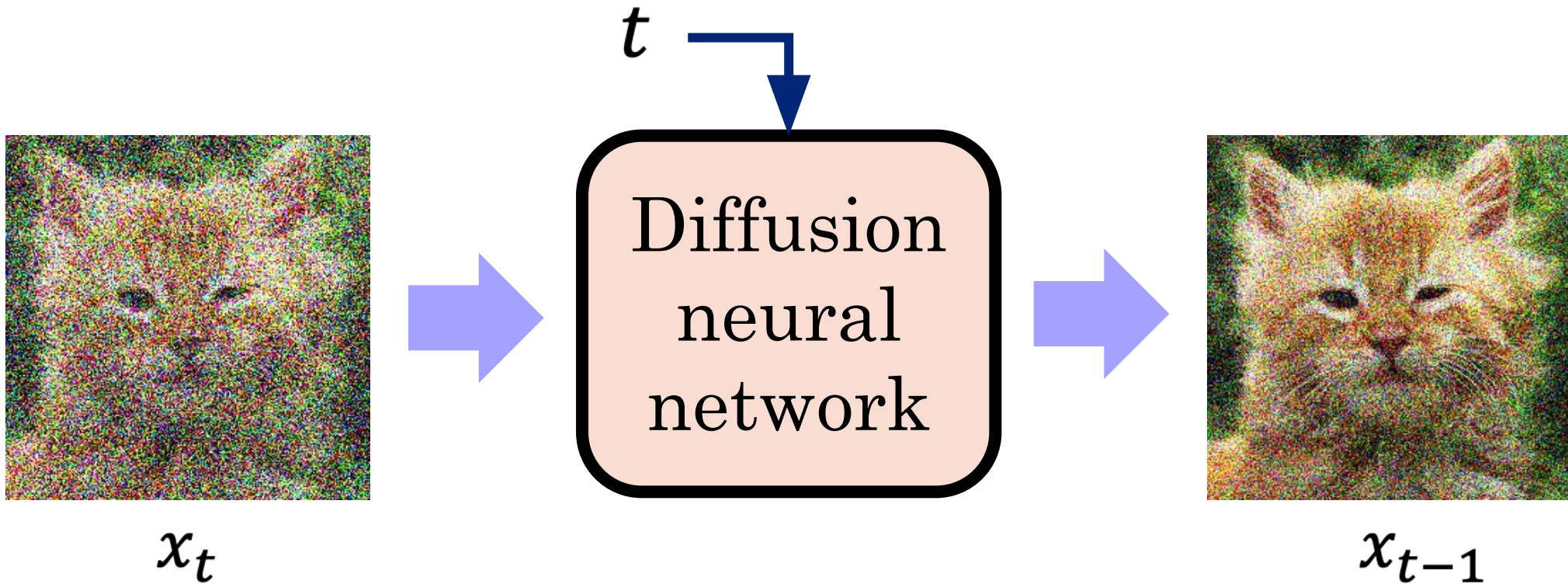
“Walking from random images towards the manifold of natural images”



Noise 2

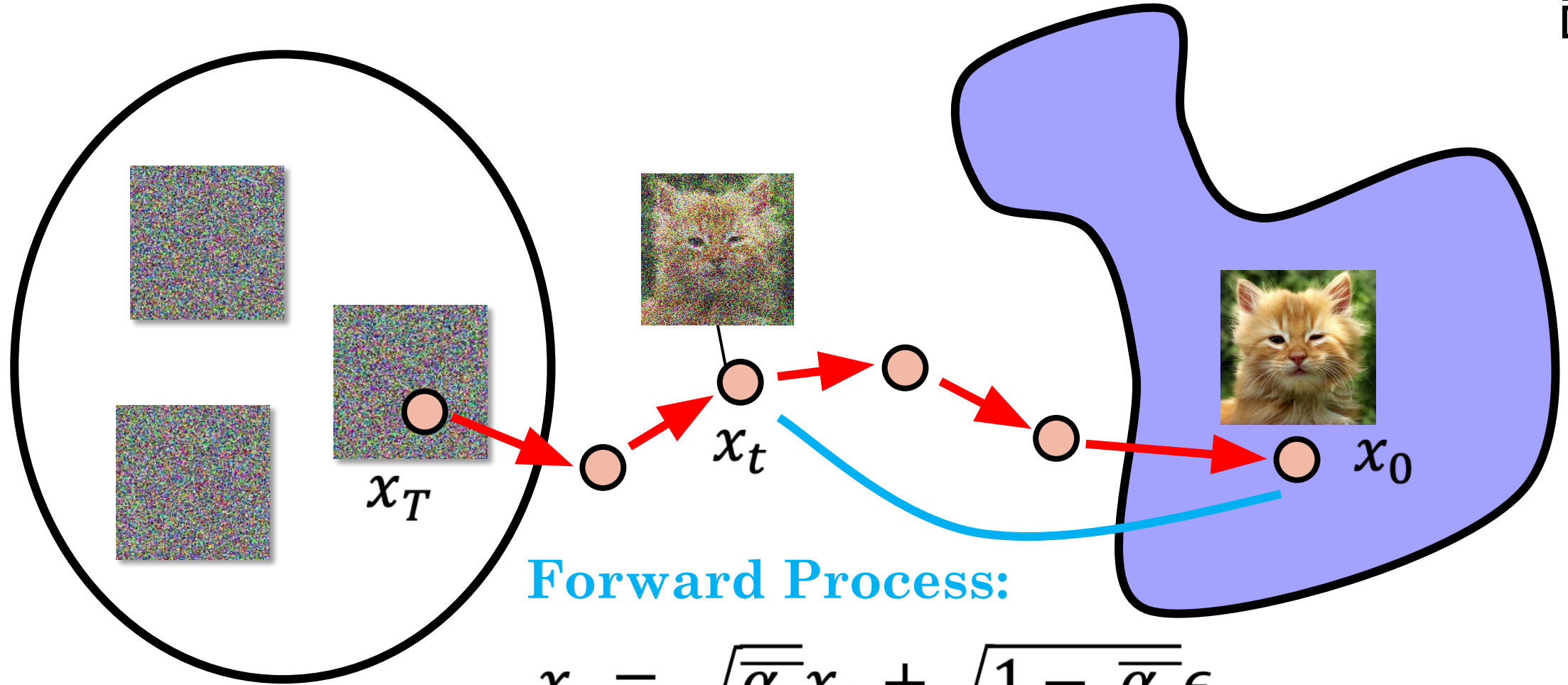
Output image 2

Denoising Diffusion Neural Network





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Forward Process:

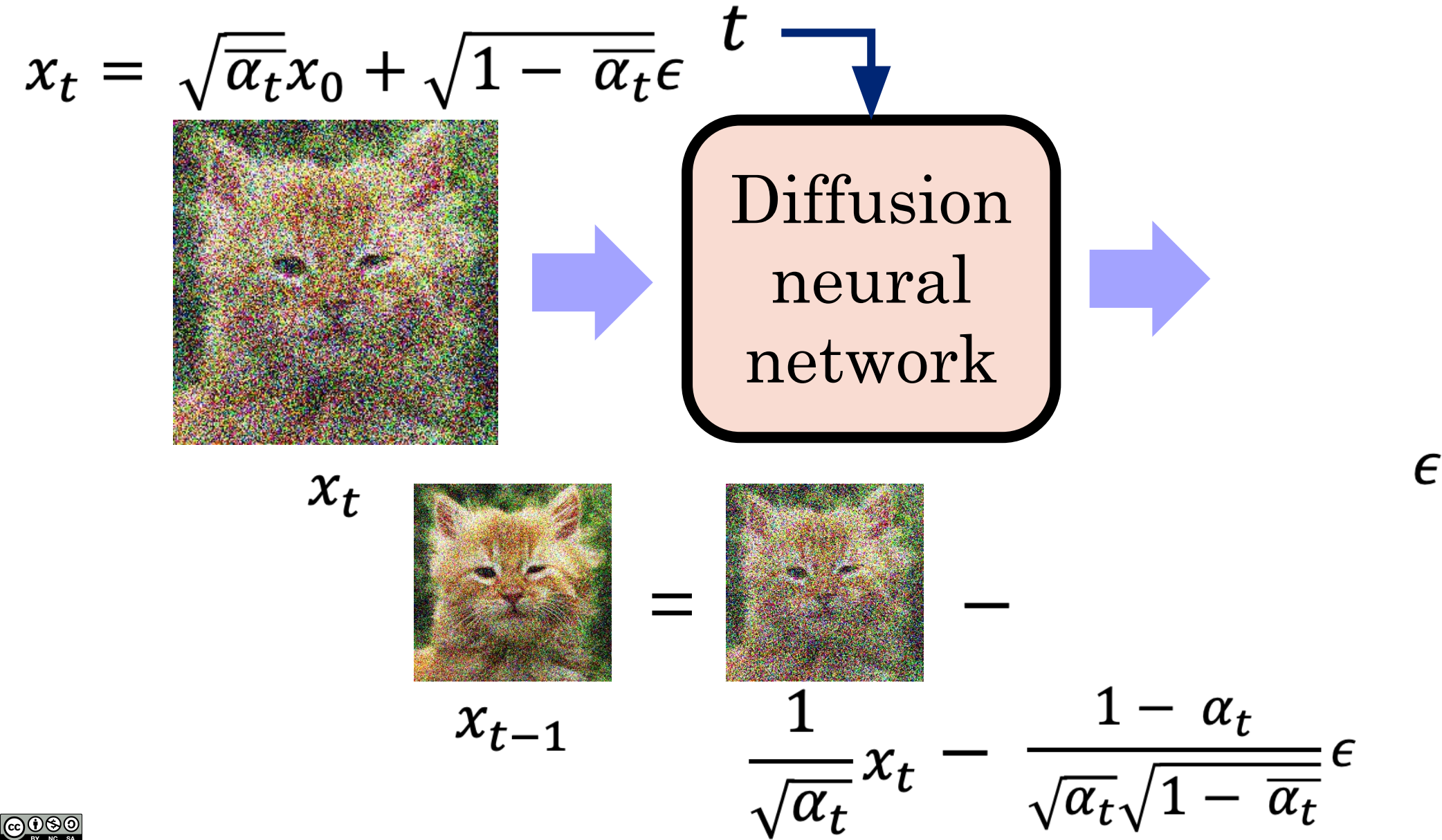
$$x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon$$

Random
images
Noise coefficient
per time step

$\epsilon \sim \mathcal{N}(0,1)$ Gaussian noise

Slide concept: Steve

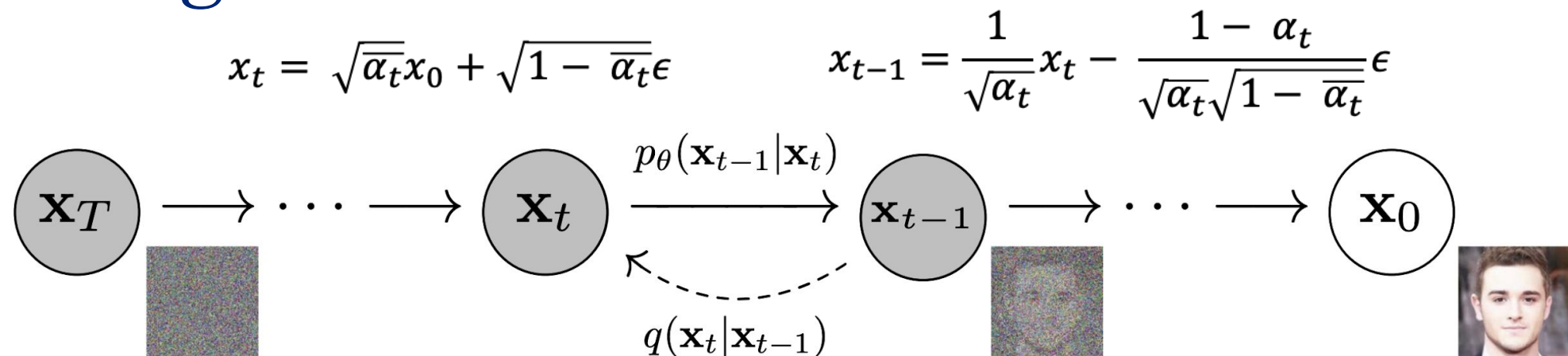
In Practice We Predict the Noise



Training a Diffusion Model



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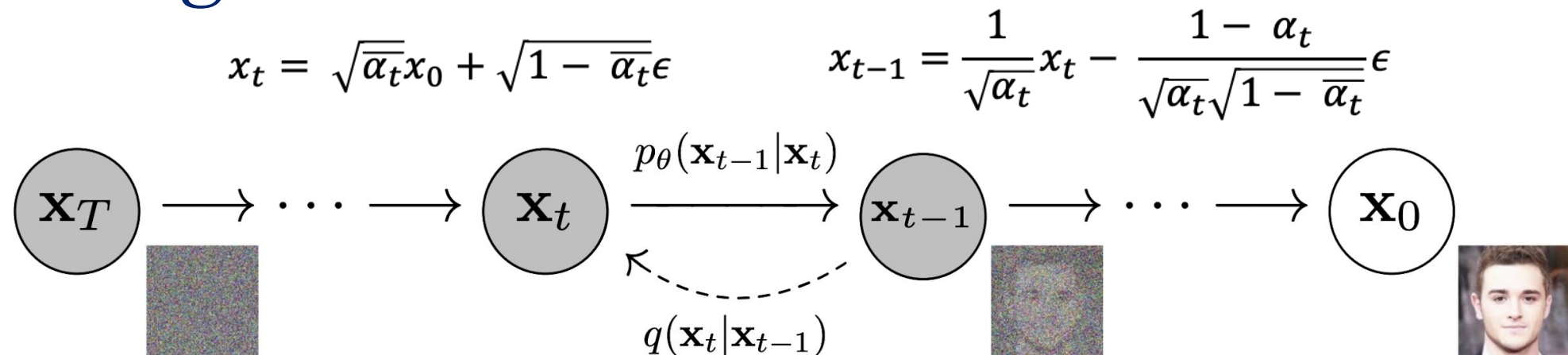
Algorithm 1 Training

- 1: **repeat**
 - 2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
 - 3: $t \sim \text{Uniform}(\{1, \dots, T\})$
 - 4: $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
 - 5: Take gradient descent step on
$$\nabla_\theta \left\| \epsilon - \epsilon_\theta(\sqrt{\alpha_t}\mathbf{x}_0 + \sqrt{1 - \alpha_t}\epsilon, t) \right\|^2$$
 - 6: **until** converged
-

Training a Diffusion Model



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Algorithm 1 Training

1: **repeat**

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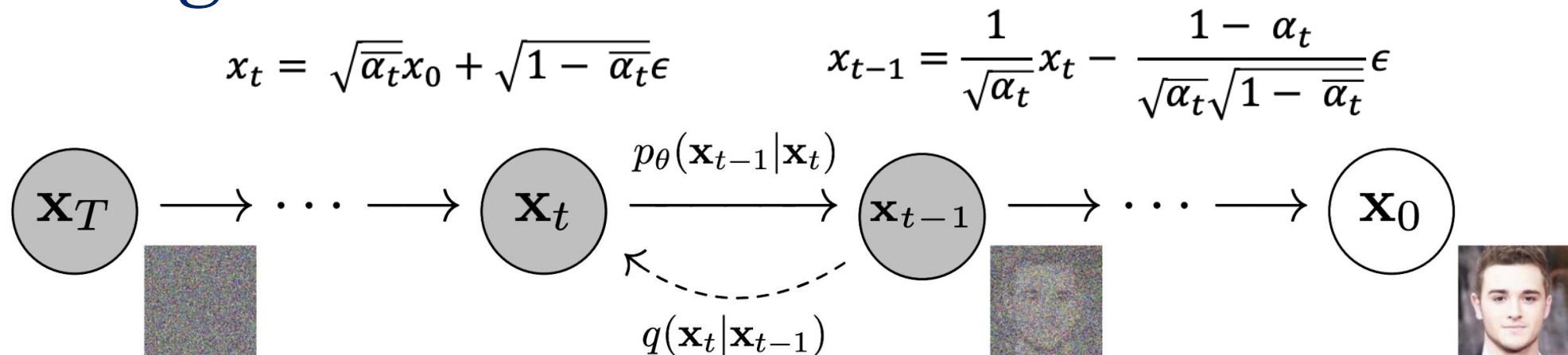
6: **until** converged

Sample x_t by adding noise to the clean image x_0

Training a Diffusion Model



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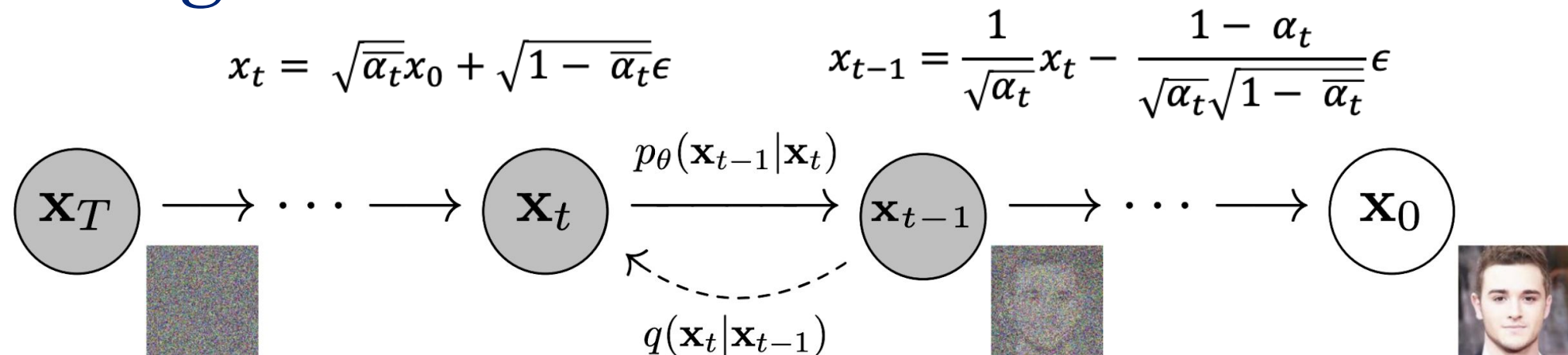
Algorithm 1 Training

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$$\nabla_{\theta} \left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2$$
- 6: **until** converged

Training a Diffusion Model



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Algorithm 1 Training

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$$\nabla_\theta \left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2$$

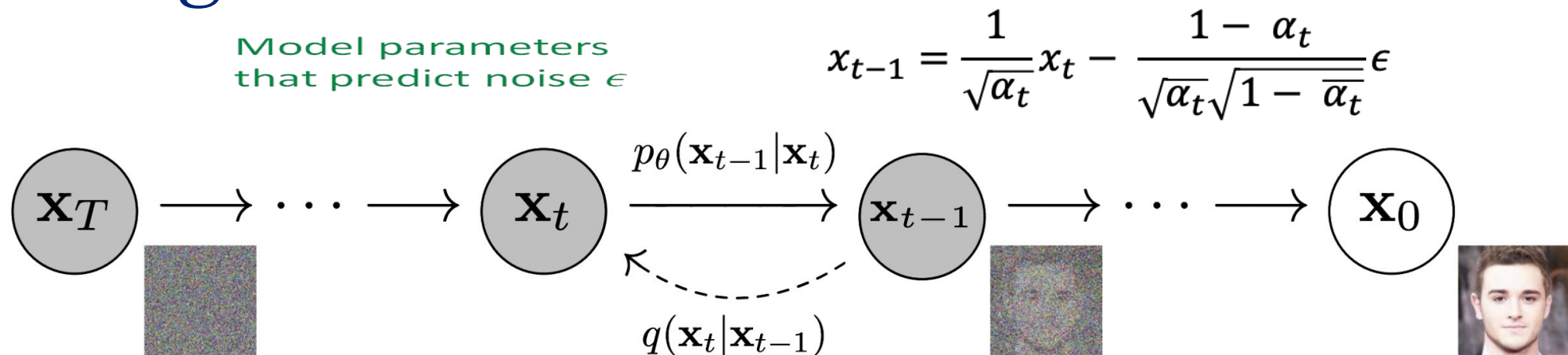
6: **until** converged

Noisy
image

Training a Diffusion Model



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Algorithm 1 Training

1: **repeat**

2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$

3: $t \sim \text{Uniform}(\{1, \dots, T\})$

4: $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

5: Take gradient descent step on

$$\nabla_{\theta} \left\| \epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2$$

6: **until** converged

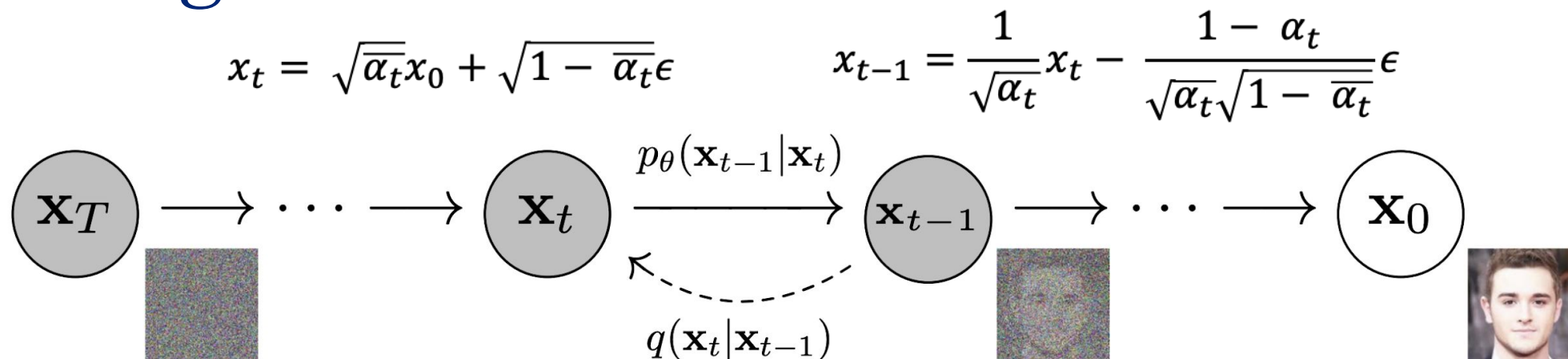
Model parameters that predict noise ϵ

Noisy image

Training a Diffusion Model



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Algorithm 1 Training

- 1: **repeat**
- 2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
- 3: $t \sim \text{Uniform}(\{1, \dots, T\})$
- 4: $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 5: Take gradient descent step on

$$\nabla_{\theta} \left\| \epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2$$

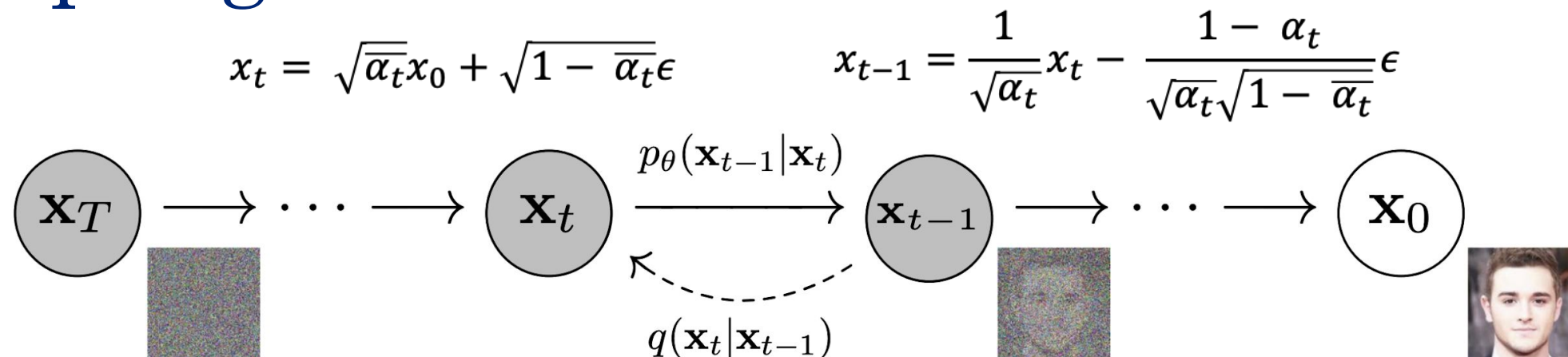
- 6: **until** converged

Make the diffusion model better at de-noising the image.

Sampling from a Diffusion Model



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Algorithm 2 Sampling

- 1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 2: **for** $t = T, \dots, 1$ **do**
- 3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
- 4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
- 5: **end for**
- 6: **return** $\mathbf{x}_0$

Remove a little bit
of noise

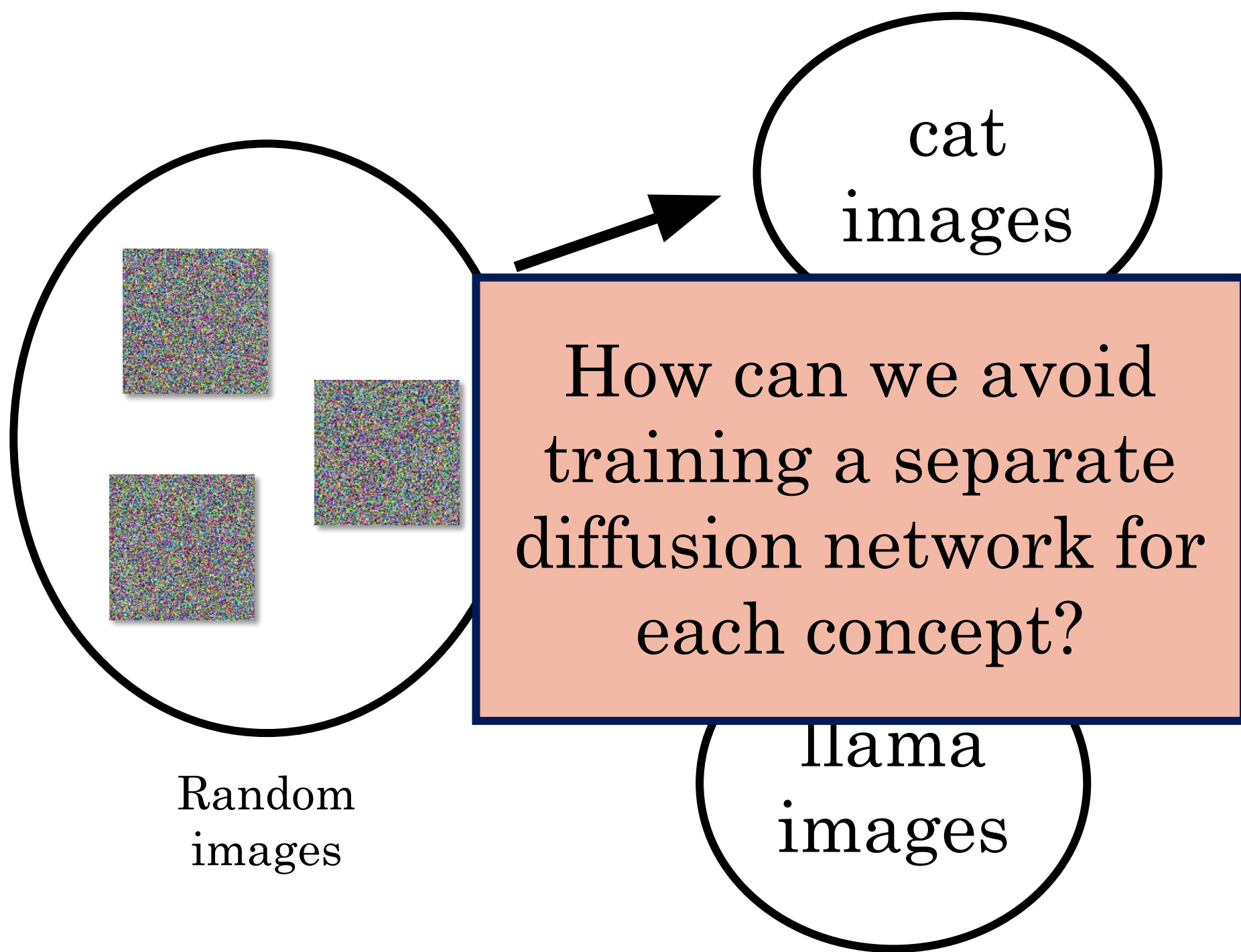
Add back a bit of noise
to avoid converging to
average



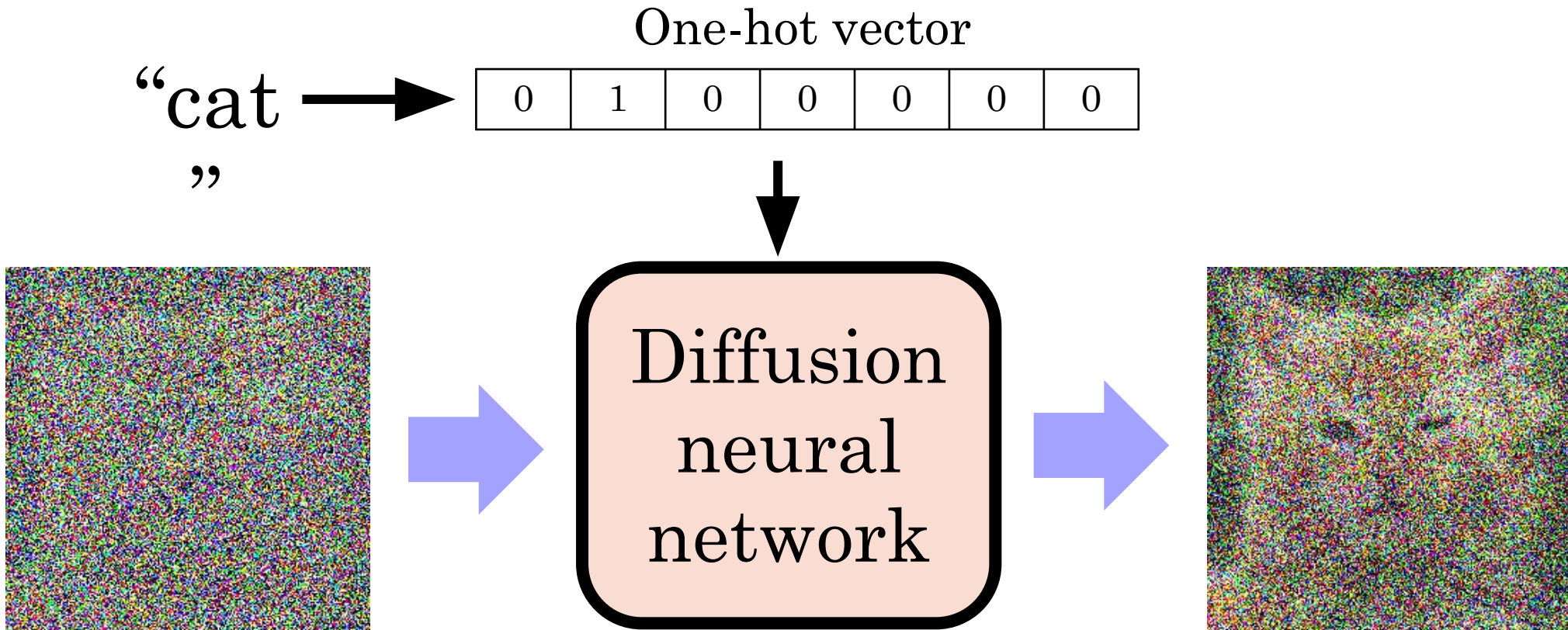
**Why does the denoising
process take many small steps
instead of a few big steps?**



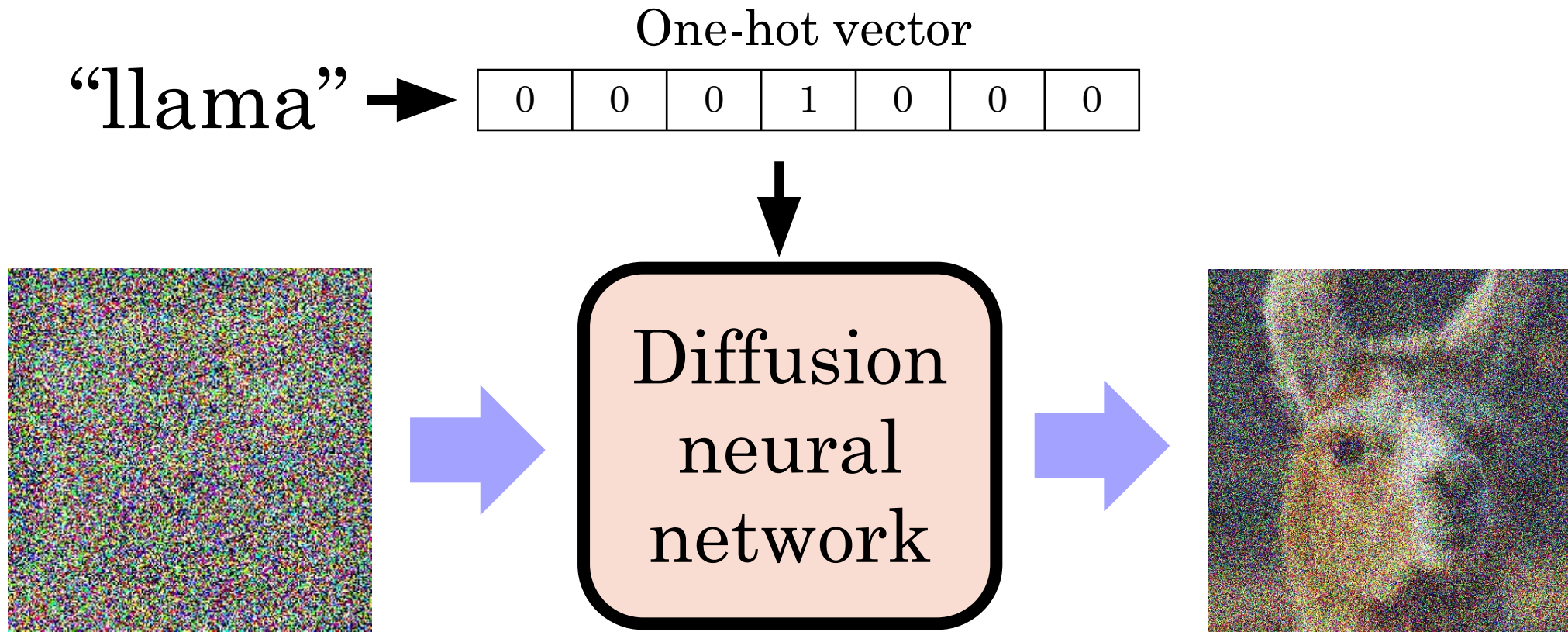
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Idea 1: Add a Class Label as Conditioning

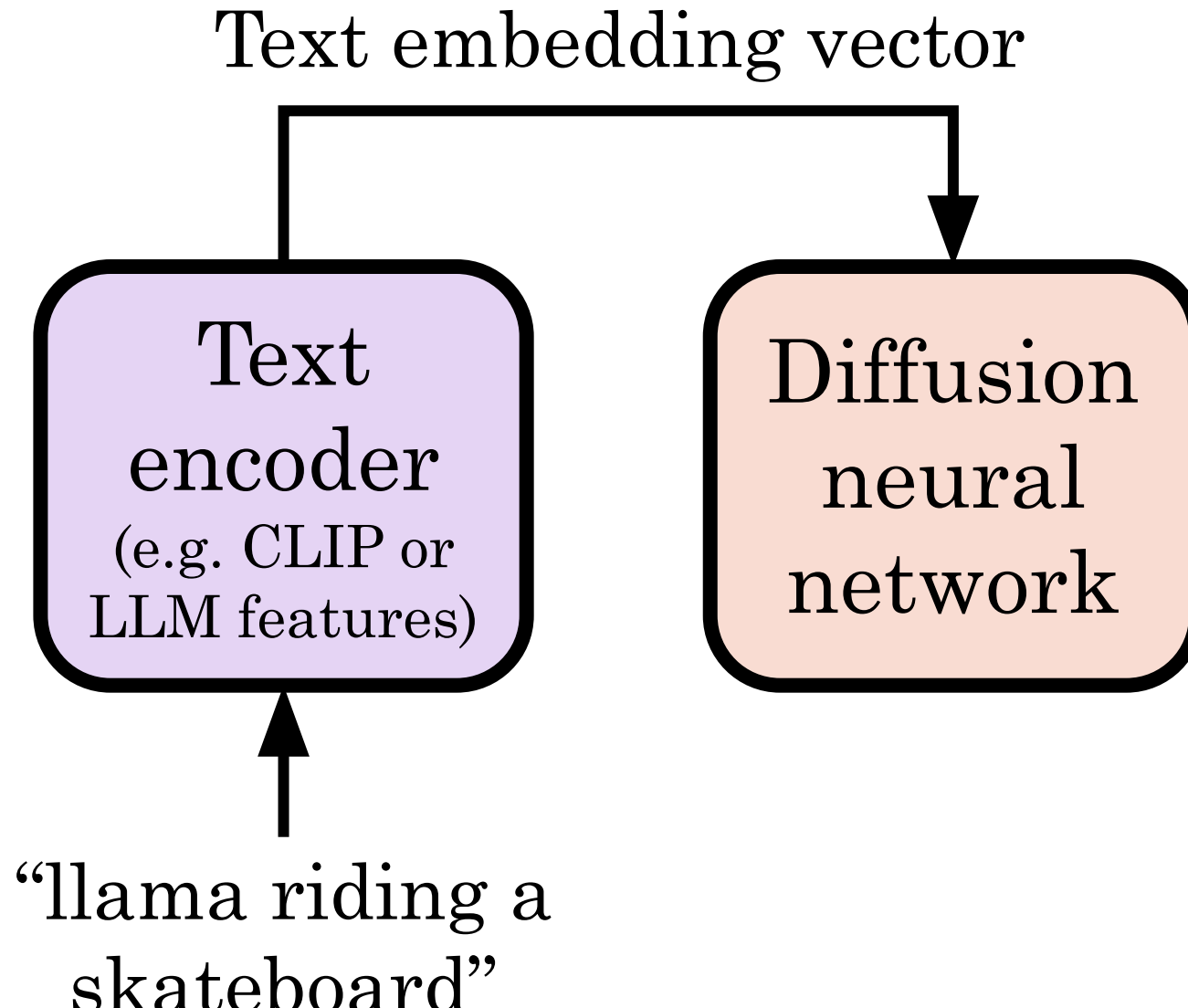


Idea 1: Add a Class Label as Conditioning

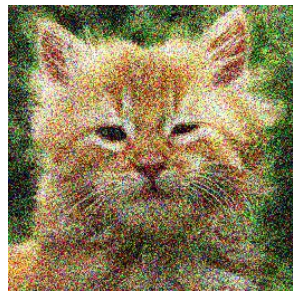


"Class conditioning" only allows for generating a fixed number of classes.

Idea 2: Add a Text Prompt as a Condition



Idea 3: Use an Image Classifier to Guide the Denoising Process ("Classifier Guidance")



x_{t-1}

=



-

+

$$\frac{1}{\sqrt{\alpha_t}} x_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t} \sqrt{1 - \alpha_t}} \epsilon$$

Update image
that makes the
input look more
like a cat

Idea 3: Use an Image Classifier to Guide the Denoising Process ("Classifier Guidance")



Update image
that makes the
input look more
like a cat

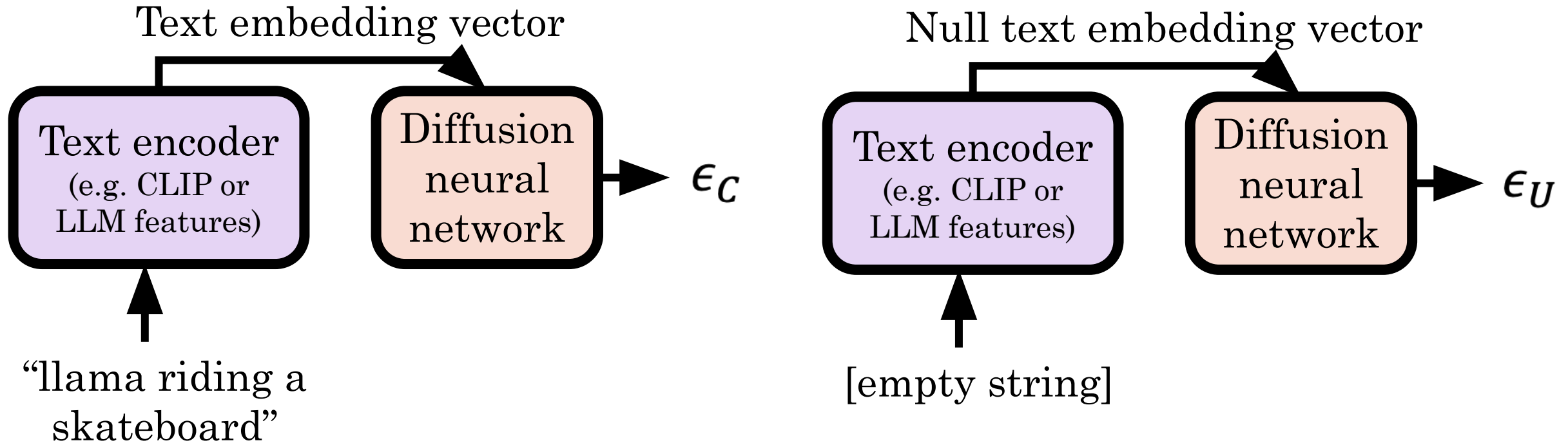
$$= \nabla F_{\theta}(x_t | y = \text{"cat"})$$

Image classifier such as
AlexNet trained on
ImageNet but
fine-tuned on noisy
images

We need to train on
noisy images and it
requires a separate
network and additional
memory

$$\tilde{\epsilon}(x_t, t, y) = \epsilon_{\theta}(x_t, t) - \gamma \nabla_{x_t} \log p(x_t | y)$$

Idea 4: Classifier-Free Guidance (CFG)

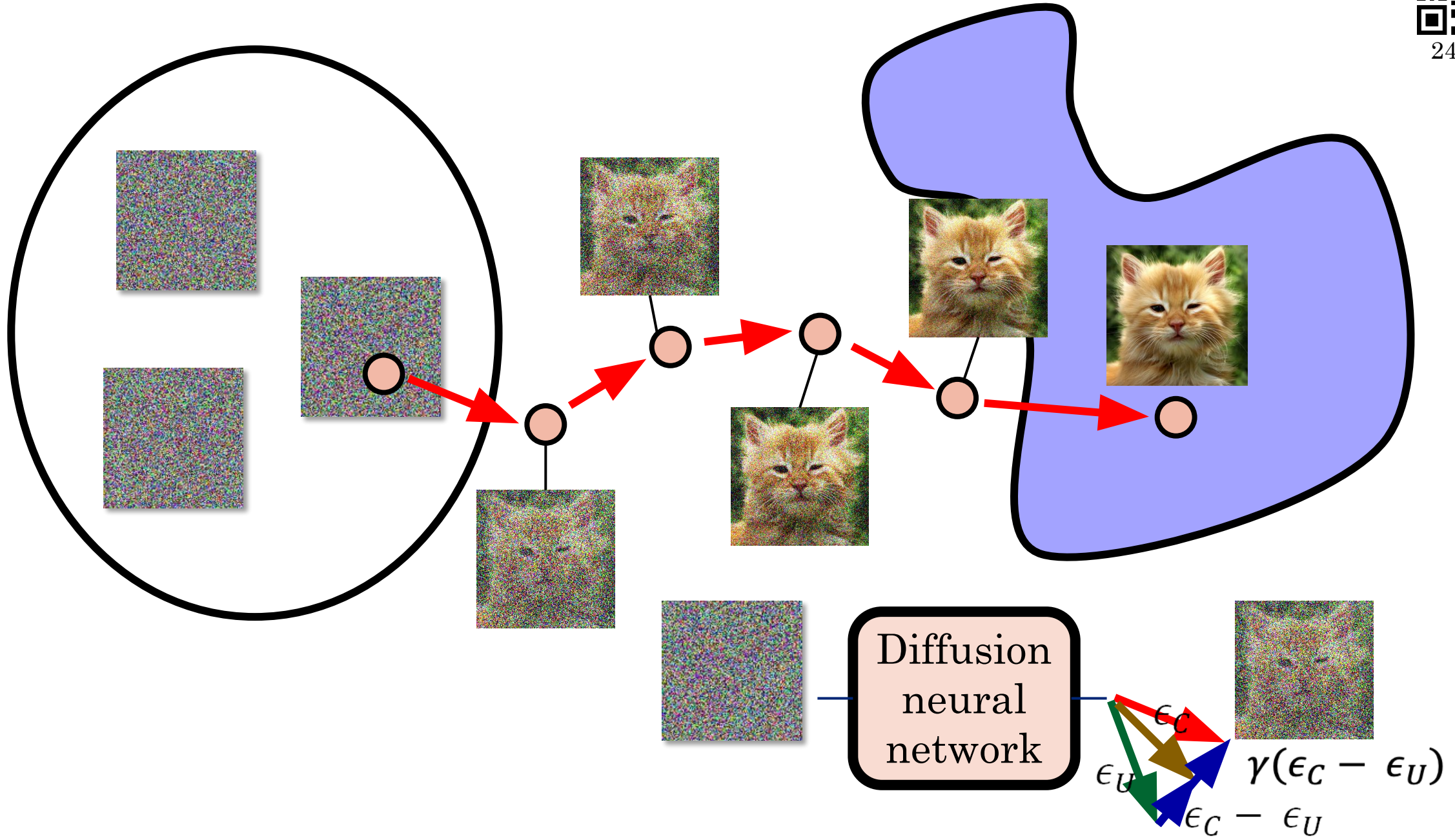


Mixed conditional and unconditional noise:

$$\epsilon = \epsilon_U + \gamma(\epsilon_C - \epsilon_U) \text{ for } \gamma \geq 0$$



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Classifier-Free Guidance (CFG)



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$\gamma = 0$



High γ

Non-guided
samples

$$\epsilon = \epsilon_U + \gamma(\epsilon_C - \epsilon_U) \text{ for } \gamma > 1$$

Classifier-free diffusion guidance: <https://arxiv.org/pdf/2207.12598>

Curious Property of Diffusion



- We are training the model to reconstruct the training set
 - But it fails!
 - Instead, it generate novel images
 - Which is what makes it great
- Perhaps it models images as textures
 - Keeping important correlations and throwing away the rest
 - But we don't know the “model space” of these textures

Lecture 25

Contrastive Learning and Diffusion Models

Reading: Partially covered in Chapter 20 of Bishop Deep Learning Textbook